A Guide to Derivations in High School Physics
Volume 1: Mechanics

Physics Help Right Now

March 10, 2026

Preface

Oftentimes, equations are given to students on various equation sheets on physics tests. Although knowing the equations is important, it is even more important to know the derivations of the equations as it provides a deeper understanding of the physical concepts behind the equations and builds physical intuition required to solve problems. Using the mathematical flexibility gained through proving formulas, students will better be able to apply their knowledge to solve more complex physics problems.

These notes attempt to derive as many formulae and concepts as possible that pertain to high school physics, from AP courses to Physics Olympiads. Equations featured in AP Physics 1 and 2 are derived without calculus (as much as possible), while calculus is employed when necessary to derive more difficult expressions. Vectors are represented using boldface ($\mathbf{v}$) in this text, and, a star (★) is used to mark the conclusion of each concept/derivation.

Before we get into it, I also want to share my experience when I learned some of the concepts in these notes for the first time. I, like most other people on this planet, can't immediately conceptualize any topic that's thrown at me. I always takes multiple revisions over weeks or even months to actually get something into my brain. Take, for example, simple harmonic motion. When I first learned the topic, I could derive the period of a pendulum and solve for its period given a few parameters. When I reviewed the topic the second time, I became comfortable solving weirder systems, like a ball bobbing water or water sloshing in a U-tube. By the third revision (about an entire year after the first!) I finally understood damping and driven oscillations. All this is to say that learning is anything but linear. If you're stuck, take a break, come back, and oftentimes the thing you were stuck on becomes trivial. If you're still stuck, reach out to a teacher. If you don't have that luxury, I'm happy to help via email (physicshelprn63@gmail.com) or 1-on-1 tutor session. Just whatever you do, try to keep an open mind, even when you're frustrated at physics for being hard, at the problem you can't solve for being annoying, and at yourself for being stuck (trust me, anybody who learned a bit of physics as been through this).

Another thing to know about me: I'm not a math super genius. I'm passionate about physics, not math, which meant that while reading through harder concepts in some textbooks I found myself spending more time working through math than understanding physics! Thus, I've tried my best to show intermediate steps in mathematical derivations and transformations to make things a bit more readable. Still, you should derive everything by yourself after reading through, since you always learn more by doing :)

The sections required for various tests are marked with the following colors:

- ▶ AP Physics 1
- ▶ AP Physics 2
- ▶ AP Physics C
- ▶ F=ma Exam

- ▶ USAPhO Exam

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Chapter 1

Mechanics

1.1 Kinematics

1.1.1 ▶ ▶ ▶ ▶ The Kinematics Equations

These five equations, collectively known as the "big five," describe all kinematic motion with constant acceleration. Calculus is often employed to derive the equations, but an algebraic derivation is provided here since these equations are required for AP Physics 1.

To derive the equations, we start with the following definitions:

$$\Delta\phi = \phi_f - \phi_i \text{ for any quantity } \phi$$

$$\phi_{avg} = \frac{\phi_i + \phi_f}{2} \text{ for any quantity } \phi$$

$$v_{avg} = \frac{\Delta x}{\Delta t}$$

$$a_{avg} = \frac{\Delta v}{\Delta t}$$

Consider a particle moving kinematically (with constant acceleration). It's position is represented by x, its velocity v, and its acceleration a. Using the first and last of the four definitions as well as the fact that acceleration is constant[1], we have:

$$\boxed{v_f = v_i + a\Delta t}. \tag{1.1}$$

Using our second definition on our third definition yields:

$$\boxed{\Delta x = \left(\frac{v_i + v_f}{2}\right)\Delta t}. \tag{1.2}$$

Rearranging eq. 1.2 and plugging it back into 1.1 gives us:

$$\Delta t = \Delta x \left(\frac{2}{v_f + v_i}\right)$$

$$v_f - v_i = \frac{2a\Delta x}{v_f + v_i}$$

$$\boxed{v_f^2 - v_i^2 = 2a\Delta x} \tag{1.3}$$

Plugging 1.1 into 1.2 gives:

$$\Delta x = \frac{1}{2}\Delta t\,(2v_i + a\Delta t)$$

$$\boxed{\Delta x = v_i\Delta t + \tfrac{1}{2}a(\Delta t)^2} \tag{1.4}$$

Lastly, we can substitute v_i from eq. 1.1 into eq. 1.4 to obtain:

$$\Delta x = (v_f - a\Delta t)\Delta t + \tfrac{1}{2}a(\Delta t)^2$$

$$\boxed{\Delta x = v_f\Delta t - \tfrac{1}{2}a(\Delta t)^2} \tag{1.5}$$

[1] i.e. $a_{avg} = a$

Thus, we have the "Big Five" equations, describing all motion with constant acceleration:

$$v_f = v_i + a\Delta t \tag{1.6}$$

$$\Delta x = \left(\frac{v_i + v_f}{2}\right)\Delta t \tag{1.7}$$

$$v_f^2 - v_i^2 = 2a\Delta x \tag{1.8}$$

$$\Delta x = v_i\Delta t + \tfrac{1}{2}a(\Delta t)^2 \tag{1.9}$$

$$\Delta x = v_f\Delta t - \tfrac{1}{2}a(\Delta t)^2 \tag{1.10}$$

★

1.1.2 ▶ ▶ ▶ ▶ The Range Equation

The range equation provides us with a way to quickly calculate the range of a projectile when launched at a certain angle θ and a certain speed v_0 on a flat surface.

To derive this equation, we start by finding the total time of flight of the projectile. We can use the first of the "Big Five" (1.6) to do this by considering vertical motion.

$$0 = v_0 \sin\theta - g t_{up} \tag{1.11}$$

$$t_{total} = 2t_{up} = 2 \cdot \frac{v_0 \sin\theta}{g} \tag{1.12}$$

where we have used the fact that the total time is twice the time the projectile takes to reach its apex due to the symmetry of the motion. We can find the range by simply multiplying the total time by the horizontal speed $v_0 \cos\theta$ since there is no horizontal acceleration. Thus,

$$\begin{aligned} R = & \frac{2v_0 \sin\theta}{g} \cdot v_0 \cos\theta \\ = & \boxed{\frac{v_0^2 \sin 2\theta}{g}} \, . \end{aligned} \tag{1.13}$$

★

1.1.3 ▶ ▶ The Projectile Trajectory Equation

All trajectories of objects in projectile motion are parabolas. Sometimes, it is useful to know the equation of these trajectories, as they can give us valuable insight into the maximum range of a projectile or the envelope of safety for a projectile (we will go over the former of these examples since the uses of this equation are not as immediately apparent as other equations derived here)[2].

We start by considering a projectile launched with a speed of v_0 at an angle of θ above the horizontal at the origin and at $t = 0$. At a certain time t, the x coordinate of this projectile is

$$x = v_0 \cos\theta \cdot t \tag{1.14}$$

since $x = v_x t$, and the y coordinate is

$$y = v_0 \sin\theta \cdot t - \tfrac{1}{2} g t^2 \tag{1.15}$$

using eq. 1.4. Substituting t from eq. 1.14 into eq. 1.15, we have

$$\boxed{y = \tan\theta \cdot x - \frac{g \sec^2\theta}{2v_0^2} \cdot x^2}. \tag{1.16}$$

The trick to use this equation is rewriting the $\sec^2\theta$ as $1 + \tan^2\theta$ using a Pythagorean identity. Making this substitution yields the following quadratic in $\tan\theta$:

$$-\frac{gx^2}{2v_0^2}(1 + \tan^2\theta) + x\tan\theta - y = 0$$

$$1 + \tan^2\theta - \frac{2v_0^2}{gx}\tan\theta + \frac{2v_0^2 y}{gx^2}$$

$$\tan^2\theta - \frac{2v_0^2}{gx}\tan\theta + \frac{2v_0^2 y}{gx^2} + 1 = 0 \tag{1.17}$$

Where the expression has been divided through to remove any coefficient in front of the $\tan^2\theta$ term.

Since this is a quadratic equation, it can be maximized by setting the discriminant of the quadratic formula to zero. Using this optimization, we can find the maximum range R_{max} of a projectile launched at *any* height H.

Let's say that a projectile is launched from a height H at an optimal angle θ_{max} such that it travels a horizontal distance R_{max}. If we plug all of these values into eq. 1.17, we get [3].

$$\tan^2\theta - \frac{2v_0^2}{gR_{max}}\tan\theta - \frac{2v_0^2 H}{gR_{max}^2} + 1 = 0. \tag{1.18}$$

[2] Although not obvious, the use of this technique has been featured multiple times on physics competitions and thus is important :)

[3] Note that the projectile travels a distance $-H$ in the vertical direction, so $y = -H$

Since the values are all optimized, the discriminant ($\sqrt{b^2 - 4ac}$) must equal zero. Writing this condition, we have

$$\frac{4v_0^4}{g^2 R_{max}^2} - 4\left(1 - \frac{2v_0^2 H}{g R_{max}^2}\right)$$

$$\frac{v_0^4}{g^2} = R_{max}^2 - \frac{2v_0^2 H}{g}$$

$$R_{max}^2 = \frac{v_0^2}{g}\left(\frac{v_0^2}{g} + 2H\right). \tag{1.19}$$

We can find θ_{max} using the quadratic formula, and since the discriminant is zero, the quadratic formula reduces down to $\tan\theta_{max} = -b/2a$. So,

$$\tan\theta_{max} = \frac{v_0^2}{gR_{max}}. \tag{1.20}$$

If we set $H = 0$ in eq. 1.18 and 1.19 (just a projectile launched on a flat surface), we get $R_{max} = v_0^2/g$ and $\theta_{max} = 45°$ as predicted in eq. 1.13 ★

1.2 Dynamics

1.2.1 ▶ ▶ ▶ ▶ Extension of $F = ma$

This rather short section concerns the full extended Newton's second law. It adds another layer of nuance that is important to note while considering Newtonian Mechanics.

The full, most rigorous representation of Newton's second law is the following:

$$\boxed{\mathbf{F}_{Net,External} = M\mathbf{a}_{cm}}. \tag{1.21}$$

Where M is the total mass of the system and $\mathbf{a}_{cm}$ is the acceleration of the center of mass of the system. To derive this equation, we first start with a system of particles. If we look at a single particle with mass m_i in this system, writing out Newton's second law gives

$$\mathbf{F}_{i,ext} + \mathbf{F}_{i,int} = m_i\mathbf{a}_i \tag{1.22}$$

where $\mathbf{F}_{i,ext}$ and $\mathbf{F}_{i,int}$ are the forces acting on the particle external and internal to the system respectively. Summing over all particles gives

$$\sum_i \mathbf{F}_{i,ext} + \sum_i \mathbf{F}_{i,int} = \sum_i m_i\mathbf{a}_i$$

$$\mathbf{F}_{Net,Ext} = M\mathbf{a}_{cm}. \tag{1.23}$$

The first term summed over the system becomes the net external force. The second term is equal to zero because of Newton's third law. The right hand side of the equation becomes $M\mathbf{a}_{cm}$ due to the definition of the center of mass i.e. $M\mathbf{a}_{cm} = \sum_i m_i\mathbf{a}_i$.

This equation states that the net force external to any system (rigid or non-rigid) equals the total mass of the system multiplied by the acceleration of the center of mass of the system. A lot of the time, it will be simply written as $F = ma$, but it is important to remember the full form of the equation ★

1.2.2 ▶ ▶ ▶ ▶ $F = dp/dt$

This section concerns the definition of force as the rate of change of momentum. The concept is described in three ways, and each section is marked with the appropriate color to denote its nuance.

▶ First, let us consider the most famous representation of Newtons Second Law for a closed system. Namely, that

$$\mathbf{F}_{net} = m\mathbf{a}. \tag{1.24}$$

Next, we consider the classical definition of linear momentum

$$\mathbf{p} = m\mathbf{v}. \tag{1.25}$$

Next, let us consider the rate of change of the linear momentum $\mathbf{p}$.

$$\frac{\Delta \mathbf{p}}{\Delta t} = m\frac{\Delta \mathbf{v}}{\Delta t} = m\mathbf{a} = \mathbf{F} \tag{1.26}$$

Thus, the force on an object equals the rate of change of its momentum.

▶ ▶ Now, let us use calculus to expand on the previous derivation. Again, we take the rate of change of the linear momentum of a particle.

$$\mathbf{F} = \frac{dp}{dt} = \frac{d}{dt}(m\mathbf{v}) = \frac{dm}{dt}\mathbf{v} + \frac{d\mathbf{v}}{dt}m \tag{1.27}$$

Where the product rule has been used, because we are no longer assuming that the system is closed (i.e. mass can enter and exit). We see that this adds a correction term to $\mathbf{F} = m\mathbf{a}$. Hence, we see that F must not always equal ma, which is in fact the case in some systems.

▶ Now, let us add yet another layer of nuance. We know from special relativity that the force applied on an object is equal to

$$\mathbf{F} = \gamma^3 m\mathbf{a}. \tag{1.28}$$

We get this equation by using $F = dp/dt$ on the equation for relativistic momentum. The significance of this equation is the fact that $F = ma$ is an approximation for small velocities and is not a fundamental "law" ★

1.2.3 ▶ ▶ ▶ ▶ Centripetal Acceleration

When an object moves in a circle at a constant velocity, it experiences an inward acceleration due the direction of the velocity vector changing with time. In this section, we derive this acceleration known as the centripetal acceleration.

First, consider a particle moving in a circle from a point A to a point B.

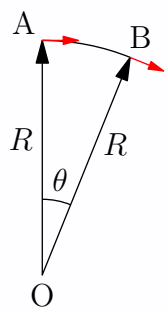


Fig. 1.1

Where the red vectors are the velocity vectors of the particle at points A and B. Connecting the tails of these two vectors together, we can find the change in velocity and thus the acceleration of this particle.

(a) Figure a **(b)** Figure b

Fig. 1.2

For small angles θ, Fig.1.2b simply becomes a triangle[4], giving us a pair of similar triangles. Thus, we have

$$\frac{v}{\Delta v} = \frac{R}{S}. \tag{1.29}$$

Using $S = v\Delta t$, we have

$$\frac{v}{\Delta v} = \frac{R}{v\Delta t}$$

$$\frac{\Delta v}{\Delta t} = \frac{v^2}{R}$$

$$\boxed{a_c = \frac{v^2}{R}}. \tag{1.30}$$

Which is the magnitude of the centripetal acceleration of an object moving in a circle ★

[4]See A.2.1 for a more rigorous proof

1.2.4 ▶ ▶ Accelerated Reference Frames

An inertial reference frame is defined as one in which Newton's laws hold. Thus, we can use Newton's laws to describe motion in all inertial reference frames. However, we sometimes want to describe motion in non-inertial reference frames, such as an accelerating car or a spinning merry-go-round. To describe motion in such frames, we need to make a small modification to Newton's description of motion using forces.

First, consider two particles A and B. A is in an inertial reference frame and B does not move with respect to A. B is inside a box moving with acceleration a to the right.

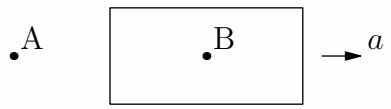


Fig. 1.3

Taking the reference frame of particle A, B is not accelerating and only the box accelerates at a. However, if we take the reference frame of the box instead of particle A, both B and A accelerate to the left with acceleration a.

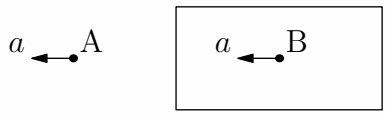


Fig. 1.4

Thus, we see that in the reference frame of the box, B has a leftward acceleration of magnitude a. By Newton's second law, it must have a force applied on it with magnitude a times its mass. The direction should be in the same direction as its acceleration, i.e. to the left. Hence, in the non-inertial reference frame of the box, particle B experiences a "force" opposite to the direction of the acceleration of the reference frame. This sort of force which exists in non-inertial reference frames is called a *fictitious* force because it doesn't exist in some inertial frames. However, it is very much a real force in non-inertial frames. For example, astronauts on the ISS are still well within the Earth's gravitational pull, but experience no net force (weightlessness) due to the fictitious centrifugal force from their orbit balancing gravity out perfectly ★

1.2.5 ▶ ▶ Reduced Mass Approach

In this section, we consider a method to approach a two-body system by reducing it to a singly body problem. Start by considering two bodies of masses m_1 and m_2 which interact via some interaction force F. The accelerations of each body are

$$a_1 = \frac{F}{m_1}$$

$$a_2 = -\frac{F}{m_2}$$

Where the signs are opposite because of Newton's third law. If we transform into the reference frame of m_1, m_2 experiences a force of magnitude F and accelerates at the relative velocity of the two bodies a_{rel}. Where

$$a_{rel} = \frac{F}{m_1} + \frac{F}{m_2}.$$

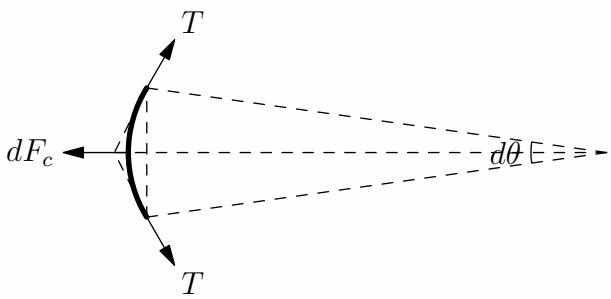
Rearranging gives

$$\begin{aligned} F &= \left(\frac{1}{m_1} + \frac{1}{m_2}\right)^{-1} a_{rel} \\ &= \mu a_{rel} \end{aligned} \tag{1.31}$$

where $\mu = \left(\frac{1}{m_1} + \frac{1}{m_2}\right)^{-1} = \frac{m_1 m_2}{m_1 + m_2}$ and is known as the reduced mass. In other words, if we transform into the reference frame of one of the bodies in a two-body system, the mass of the other body effectively becomes the reduced mass, and its motion is defined by Newton's second law ★

1.2.6 ▶ ▶ Rubber bands

Sometimes, physics Olympiads feature problems with rubber bands, so it's important to know how to approach such problems. This section addresses a relation between the different forces in a rubber band, as well as the general method to approach rubber band problems. First, consider a stretched rubber band with tension T being held in a stretched position by a radially outward force F_c. To approach any rubber band problem, consider a small angle $d\theta$ that subtends a small arc and analyze the forces on it. In this case, cutting a small arc and drawing forces on it (as well as a few guidelines) yields the following diagram



Using similar triangles, the piece of rubber band experiences a force

$$2T\sin\left(\frac{d\theta}{2}\right) \approx 2T\frac{d\theta}{2} = Td\theta,$$

to the right. To the left, the band simply experiences a force dF_c. Thus, since the band is in equilibrium, we have

$$dF_c = Td\theta$$

or simply

$$\boxed{F_c = 2\pi T} \tag{1.32}$$

★

1.3 Energy and Momentum

1.3.1 ▶ ▶ ▶ Conservative Forces

This short section shows that defining a conservative force as a derivative of a scalar potential and a force which has a circulation of zero are equivalent.

The total work done on a particle going along a path p is

$$W = \int_p \mathbf{F} \cdot d\mathbf{r} \tag{1.33}$$

One definition of a conservative force is that it is able to be written as a derivative[5] of a scalar potential, i.e.

$$\mathbf{F} = -\frac{dU}{d\mathbf{r}} \tag{1.34}$$

Taking the dot product in the integral for work yields

$$W = \int_p \mathbf{F} \cdot d\mathbf{x} = \int_p F_x dx \tag{1.35}$$

Lastly, substituting our definition of a conservative force, we have

$$W = \int_p -\frac{dU}{dx} dx = \int_p -dU \tag{1.36}$$

Taking this integral over a closed path gives zero, since the starting and ending points are the same.

$$W = \oint_p -dU = 0. \tag{1.37}$$

Eq. 1.35 the other definition of a conervative force, and thus the two definitions are equivalent ★

[5] More rigorously, gradient

1.3.2 ▶ ▶ ▶ ▶ ▶ Energy Balance & Conservation of energy

To derive the Work-Energy theorem, we start with the definition of work

$$W = F\Delta x. \tag{1.38}$$

We can substitute Δx from the kinematics equation $v_f^2 - v_i^2 = 2a\Delta x$ into this definition, as well as $F = ma$ from Newton's second law.

$$\begin{aligned} W &= ma\left(\frac{v_f^2 - v_i^2}{2a}\right) \\ &= \frac{1}{2}m(v_f^2 - v_i^2) \end{aligned} \tag{1.39}$$

Defining our kinetic energy, or energy due to motion, as $K \equiv \frac{1}{2}mv^2$, we have

$$\boxed{W = \Delta K}. \tag{1.40}$$

This equation says that the total work done on a system is equal to the change in kinetic energy of the system. However, we can further split each side of this equation into more components. The total work done on a system is a sum of external work W_{ext}, internal non-conservative work W_{nc}, and internal conservative work W_c.

$$W_{ext} + W_c + W_{nc} = \Delta K \tag{1.41}$$

Using the definition of potential energy (i.e. $W_c = -\Delta U_c$), we have

$$W_{ext} + W_{nc} = \Delta K + \Delta U_c \tag{1.42}$$

Where U_c is any sort of potential energy due to a conservative force (gravity, spring, etc.). When $W_{ext} = W_{nc} = 0$, $\Delta K + \Delta U_c = 0$, which means that energy is conserved. Next, internal non-conservative work in the system can be regarded as changes in the structure of the system, which can be accounted for by changes in chemical or thermal energy. For example, if we consider the system of a block sliding on the floor, the non-conservative work done by friction is dissipated as heat. Thus, we have

$$\boxed{W_{ext} = \Delta K + \Delta U_c + \Delta U_{th} + \Delta U_{chem}} \tag{1.43}$$

▶ ▶ From thermodynamics, we know that heat is another way to transfer energy, similar to work. Thus, our energy balance equation is

$$\boxed{Q_{ext} + W_{ext} = \Delta K + \Delta U_c + \Delta U_{th} + \Delta U_{chem}}. \tag{1.44}$$

Where any internal heat has been accounted for as changes in the thermal energy of the system ★

1.3.3 ▶ ▶ ▶ ▶ Conservation of Momentum

Momentum for a system of particles is conserved when there are no external net forces acting on the system. To derive this fact, we start by considering a system of two particles that interact via some interaction force for a time Δt.

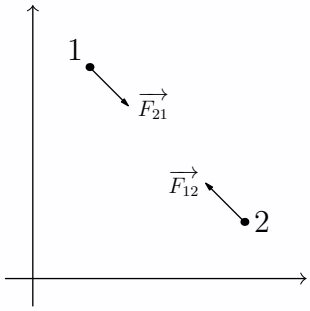


Fig. 1.5

In this diagram, $\mathbf{F_{12}}$ is the interaction force that 1 exerts on 2 while $\mathbf{F_{21}}$ is the force that 2 exerts on 1. Thus, we can calculate the impulse experienced by each particle.

$$\Delta \mathbf{p}_1 = \mathbf{F}_{21}\Delta t$$

$$\Delta \mathbf{p}_2 = \mathbf{F}_{12}\Delta t$$

Since $\mathbf{F}_{12}$ and $\mathbf{F}_{21}$ are a Newton's 3rd law pair, we have

$$\mathbf{F}_{21} = -\mathbf{F}_{12} \tag{1.45}$$

OR

$$\Delta \mathbf{p}_1 = -\Delta \mathbf{p}_2. \tag{1.46}$$

Thus, the total change in momentum of the system is

$$\sum \Delta \mathbf{p} = \Delta \mathbf{p}_1 + \Delta \mathbf{p}_2 = 0. \tag{1.47}$$

Note that we assumed that there were no external net forces to the system. If there were, the total change in momentum of each particle would not be equal in magnitude and thus momentum would not be conserved ★.

1.3.4 ▶ ▶ ▶ ▶ Relative Velocity for Elastic Collisions

When solving for the velocities in an elastic collision, we use two conservation laws: conservation of momentum, and since its an elastic collision, conservation of kinetic energy. These two conservation laws look like the following

$$m_1\mathbf{v}_{1i} + m_2\mathbf{v}_{2i} = m_1\mathbf{v}_{1f} + m_2\mathbf{v}_{2f} \tag{1.48}$$

$$\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2 \tag{1.49}$$

Because of the kinetic energy conservation equation, this system often gives a quadratic, which can often be cumbersome to solve. If we can find another equation that is linear, it would make the process of solving for the velocities of the particles much easier. Luckily, there is another equation that we can derive to use to make solving for the velocity easier. To derive this equation, we first rearrange both of the above equations to factor out the masses of the objects.

$$m_1(\mathbf{v}_{1i} - \mathbf{v}_{1f}) = m_2(\mathbf{v}_{2f} - \mathbf{v}_{2i})$$

$$m_1(v_{1i}^2 - v_{1f}^2) = m_2(v_{2f}^2 - v_{2i}^2)$$

Dividing the equations yields:

$$\boxed{\mathbf{v}_{1i} + \mathbf{v}_{1f} = \mathbf{v}_{2i} + \mathbf{v}_{2f}} \tag{1.50}$$

We can use this equation along with the conservation of momentum equation (1.47) to quickly solve for velocities in an elastic collision ★

1.3.5 ▶ ▶ ▶ Total Kinetic Energy of Any System

This section a derivation for a frequently used form of the total kinetic energy of a system by separating the kinetic energy of the system into energy due to the center of mass and energy due to relative motion. Note that the section title indicates that Physics C students should review this concept, which I deem true because the topics here involve concepts used in rotational kinetic energy and the parallel-axis theorem. However, one could very well pass Physics C without the extra nuance provided by this proof[6].

First, consider an arbitrary system (rigid or non-rigid) with total mass M in a coordinate system with center of mass at $\mathbf{r}_{cm}$. Further, consider the ith at the position $\mathbf{r}_i$ and at a position $\mathbf{r}_{icm}$ relative to the center of mass.

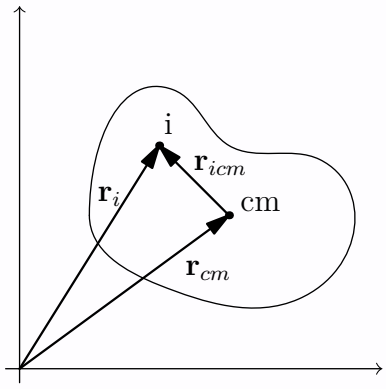


Fig. 1.6

Notice that $\mathbf{r}_i = \mathbf{r}_{cm} + \mathbf{r}_{icm}$ and thus $\mathbf{v}_i = \mathbf{v}_{cm} + \mathbf{v}_{icm}$. The kinetic energy of this particle is thus

$$\begin{aligned} K_i &= \frac{1}{2} m_i \mathbf{v}_i^2 \\ &= \frac{1}{2} m_i (\mathbf{v}_{cm} + \mathbf{v}_{icm})^2 \\ &= \frac{1}{2} m_i (v_{cm}^2 + 2\mathbf{v}_{cm} \cdot \mathbf{v}_{icm} + v_{icm}^2) \end{aligned} \tag{1.51}$$

Summing over all particles yields the following sum:

$$\begin{aligned} K_{total} = &\sum_i K_i = \\ &\sum_i \frac{1}{2} m_i v_{cm}^2 \\ &+ \sum_i m_i (\mathbf{v}_{cm} \cdot \mathbf{v}_{icm}) \\ &+ \sum_i \frac{1}{2} m_i v_{icm}^2 \end{aligned}$$

[6] F=ma and USAPhO students, though, should stay!

The first term is simply equal to $\frac{1}{2}Mv_{cm}^2$ because v_{cm} is independent of the ith particle. The second term is a bit more difficult to understand, though. Thus, prepare yourself! The next part gets quite dense. Rearranging the second term in a very specific way yields:

$$\mathbf{v}_{cm} \cdot \frac{d}{dt}\left(\sum_i m_i \mathbf{r}_{icm}\right) \tag{1.52}$$

To understand this new sum recall the definition of center of mass, namely that

$$M\mathbf{r}_{cm} = \sum_i m_i \mathbf{r}_i$$

This equation says that the product of the object's total mass M times the distance from the origin to the center of mass of a system $\mathbf{r}_{cm}$ is equal to the sum of the products of the mass of each constituent particle m_i and its distance to the origin $\mathbf{r}_i$. In eq. 1.51, however, notice that the distance in the sum $\mathbf{r}_{icm}$ is the distance of each particle from the center of mass, implying that the origin is centered on the center of mass. That means that the sum in eq. 1.51 is just equal to the total mass of the object M times the distance from the center of mass to the center of mass, which is zero!

Hence, we have

$$K_{total} = \frac{1}{2}Mv_{cm}^2 + \sum_i \frac{1}{2}m_i v_{icm}^2$$

Where v_{icm} is the relative velocity of the ith particle to the center of mass ★

1.3.6 ▶ ▶ Total Kinetic Energy of a Two-Body System

In this section, we use the result from 1.3.5 to derive an expression for the total energy of a two-body system. We specifically consider this system since it is the one under consideration for all two-body collisions.

Start by considering two masses m_1 and m_2 moving at velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively. Since we wish to find the sum $\sum \frac{1}{2} m_i v_{icm}^2$, we should find the velocity of each mass with respect to the velocity of their center of mass. By definition, the center of mass velocity is

$$\mathbf{v}_{cm} = \frac{m_1\mathbf{v}_1 + m_2\mathbf{v}_2}{m_1 + m_2}.$$

The velocity of m_1 with respect to the center of mass $\mathbf{v}_{1cm}$ is therefore

$$\begin{aligned}
\mathbf{v}_{1cm} &= \mathbf{v}_1 - \mathbf{v}_{cm} \\
&= \frac{m_1\mathbf{v}_1 + m_2\mathbf{v}_1 - m_1\mathbf{v}_1 - m_2\mathbf{v}_2}{m_1 + m_2} \\
&= \frac{m_2}{m_1 + m_2}(\mathbf{v}_1 - \mathbf{v}_2) \\
&= \frac{m_2}{m_1 + m_2}\mathbf{v}_{rel}
\end{aligned}$$

Where $\mathbf{v}_{rel} \equiv \mathbf{v}_1 - \mathbf{v}_2$ is the relative velocity of the two bodies. By symmetry, $\mathbf{v}_{2cm}$ equals

$$-\frac{m_1}{m_1 + m_2}\mathbf{v}_{rel}$$

Now, we can write the sum using these values

$$\begin{aligned}
\sum \frac{1}{2} m_i v_{icm}^2 &= \frac{1}{2} m_1 v_{1cm}^2 + \frac{1}{2} m_2 v_{2cm}^2 \\
&= \frac{1}{2}\frac{m_1 m_2^2}{(m_1 + m_2)^2} v_{rel}^2 + \frac{1}{2}\frac{m_1^2 m_2}{(m_1 + m_2)^2} v_{rel}^2 \\
&= \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2} v_{rel}^2 \left(\frac{m_1 + m_2}{m_1 + m_2}\right) \\
&= \frac{1}{2}\mu v_{rel}^2
\end{aligned}$$

Where $\mu \equiv \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass of the system. Hence, the total kinetic energy of a two-body system is equal to

$$\boxed{K_{total} = \frac{1}{2} M v_{cm}^2 + \frac{1}{2}\mu v_{rel}^2} \tag{1.53}$$

This expression also provides an easy way to find the energy lost in a perfectly-inelastic collision, since v_{rel} after the collision is zero. Thus, the total energy lost in such a collision is $\frac{1}{2}\mu v_{rel}^2$ ★

1.3.7 ▶ ▶ Obtaining Equations of Motion Using Energy

$F = ma$ gives us an important relation between the force applied on an object and its acceleration. If we can write out $F = ma$ for every object in a system, we can solve for the total movement of the system, or its equations of motion. However, finding and writing forces for every object can often be extremely tedious, which is the exact reason why conserved quantities like energy and momentum are so important. In particular, we can sometimes use energy to find the equation of motion of certain systems if a forces approach is not straightforward.

Consider an arbitrary system that depends on a certain coordinate x. the total energy of the system is

$$E = \frac{1}{2}m\dot{x}^2 + U(x) \tag{1.54}$$

where m is the mass of the system and $U(x)$ is the potential energy of the system. Since energy is conserved, $dE/dt = 0$, or

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + \frac{dU}{dx}\dot{x} = 0.$$

Using the definition of potential energy, we have

$$-\frac{dU}{dx}\dot{x} = m\ddot{x}$$

$$F = ma$$

What the above expression implies is that if we have a non-trivial system that only depends on a single coordinate, we can write the total energy and take a time derivative to find the equation of motion of the system. This process can oftentimes be a lot easier that writing out forces, such as a system of water oscillating in a U-tube, or a complicated spring oscillator.

In particular, if a system has a total energy of the form

$$E = \frac{1}{2}Ax^2 + \frac{1}{2}Bv^2$$

where A and B are constants and x and v are the position and velocity respectively, the particle oscillates with angular frequency $\boldsymbol{\omega} = \sqrt{A/B}$ ★

1.4 Rotation

1.4.1 ▶ ▶ ▶ ▶ Angular Quantities Are the Same About Any Pivot

Consider a rigid body rotating freely with angular velocity $\boldsymbol{\omega}$ about an arbitrary point 1 on the body. Then, consider another point 2 a distance r away from the point of rotation. By the definition of rigid body rotation, this point has a speed $v = r\boldsymbol{\omega}$ in the direction shown in the diagram.

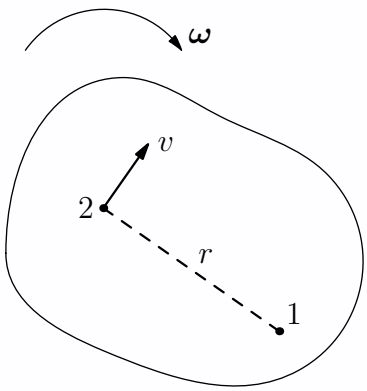


Fig. 1.7

Transforming into the reference frame of 2, we can find the angular velocity of the object about this point. Since it's a rigid body, the angular velocity of the body at 2 is equal the speed of 1 in 2's reference frame divided by r. Since 1 and 2 move relative to each other, this velocity is simply v. Hence,

$$\begin{aligned}\boldsymbol{\omega}_2 &= v/r \\ &= \frac{r\boldsymbol{\omega}}{r} \\ &= \boldsymbol{\omega}\end{aligned}$$

Thus, the angular velocity about any point on a rigid body is the same. It also follows that other angular quantities are the same, such as angular acceleration. This result is important, since the angular acceleration of a rigid body can be found by taking the torque about any fixed point, an important fact if some axes are easier to take torque around ★

1.4.2 ▶ ▶ ▶ ▶ Kinetic Energy of Rigid Bodies (Rotational KE)

▶ The following equation for rotational kinetic energy is required for AP Physics 1, but the derivation is not required. However, nuance from the derivation could be useful.

$$K_{rot} = \frac{1}{2} I \omega^2 \tag{1.55}$$

▶ ▶ ▶ To derive this formula, first consider an arbitrary rotating body (rigid or non-rigid) rotating with speed $\boldsymbol{\omega}$ and translating with speed v_{cm}.

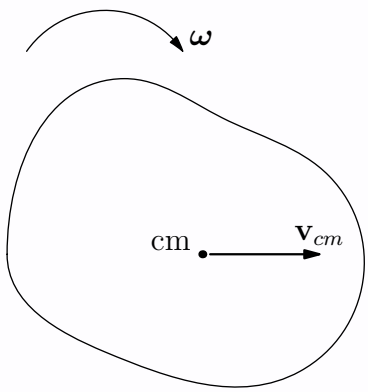


Fig. 1.8

To take the total kinetic energy of this body, we can use the formula derived in 1.3.5:

$$K_{total} = \frac{1}{2} M v_{cm}^2 + \sum \frac{1}{2} m_i v_{icm}^2 \tag{1.56}$$

Remember that this equation is always valid, but it can be difficult to calculate the sum in the second term for every system. So, the objective here is to simplify the sum for certain systems, making it easier to calculate the total energy for these systems. In 1.3.6, we derived the simplified expression for a system of two bodies, and here we derive a simplified expression for rigid bodies.

To begin, we first transform into the reference frame of the center of mass, since we need to find the velocity of the particles relative to the center of mass. Then, we can consider an arbitrary mass element on the body moving at a velocity $\mathbf{u}$ relative to the center of mass at a position $\mathbf{r}_{icm}$ relative to the center of mass. We can split this $\mathbf{u}$ vector into two components: one parallel to $\mathbf{r}_{icm}$ and one perpendicular to $\mathbf{r}_{icm}$. Thus, we have

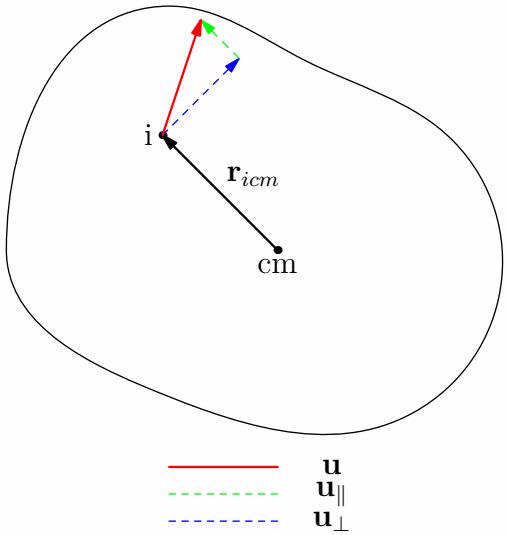


Fig. 1.9

where $\mathbf{u} = \mathbf{u}_{\parallel} + \mathbf{u}_{\perp}$. Since we are using polar coordinates centered at the center of mass, we can represent these vectors as

$$\mathbf{u} = \frac{d\mathbf{r}_{icm}}{dt} + r_{icm}\frac{d\boldsymbol{\theta}}{dt}.$$

Since we are considering a rigd body, $\frac{d\mathbf{r}_{icm}}{dt} = 0$ by definition. Thus, writing our total kinetic energy yields

$$K_{total} = \frac{1}{2}Mv_{cm}^2 + \sum \frac{1}{2}m_i r_{icm}^2 \omega^2.$$

If we define the moment of inertia about the center of mass I_{cm} to be $\sum m_i r_{icm}^2$, we have

$$\boxed{K_{total} = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2}. \tag{1.57}$$

This equation states that we can write the total kinetic energy of a rigid body as the kinetic energy due to motion of the center of mass plus the kinetic energy due to rotation about the center of mass.

What if, however, we want to consider the kinetic energy of a rigid body rotating about a fixed pivot? In that case, consider the same rigid body in Fig. 1.8, but this time fixed about point i. Thus, we will also accordingly set our coordinate system to be centered at i.

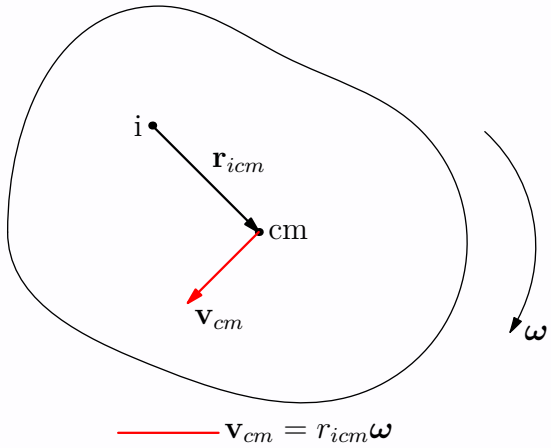


Fig. 1.10

In this case, the center of mass has some velocity $\mathbf{v}_{cm}$ relative to point i. Using the definition of a rigid body and the definition of angular velocity, we have $\mathbf{v}_{cm} = r_{icm}\boldsymbol{\omega}$. Next, since we are solving for rotation about a point separate from the center of mass, it is appropriate to use the Parallel-Axis theorem. Namely, that

$$I_i = I_{cm} + Mr_{icm}^2 \tag{1.58}$$

Plugging I_{cm} into eq. 1.56 gives

$$\begin{aligned} K_{total} &= \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}\omega^2(I_i - Mr_{icm}^2) \\ &= \frac{1}{2}Mv_{cm}^2 - \frac{1}{2}M(r_{icm}\boldsymbol{\omega})^2 + \frac{1}{2}I_i\boldsymbol{\omega}^2 \\ &= \boxed{\frac{1}{2}I_i\boldsymbol{\omega}^2} \end{aligned} \tag{1.59}$$

This expression gives the total kinetic energy of a rigid body rotating about a fixed pivot. Notice that if we transform into the center of mass reference frame here, we essentially have the same scenario as depicted in fig. 1.9.

To summarize, the total kinetic energy of a freely moving rigid body is comprised of two parts: the kinetic energy due to translation of the center of mass $\frac{1}{2}Mv_{cm}^2$ and the kinetic energy due rotation about the center of mass $\frac{1}{2}I_{cm}\omega^2$. For a rigid body rotating about a fixed pivot, the total kinetic energy is simply $\frac{1}{2}I_i\boldsymbol{\omega}^2$ where I_i is the moment of inertia about the fixed pivot ★

1.4.3 ▶ ▶ ▶ Parallel Axis Theorem

The Parallel Axis Theorem provides a way to find the moment of inertia of a rigid body about an axis parallel to an axis that passes through the center of mass. First, consider the following rigid body with moment of inertia I_{cm} about the center of mass. Further consider that we wish to find the moment of inertia about the axis perpendicular to the page and passing through point O.

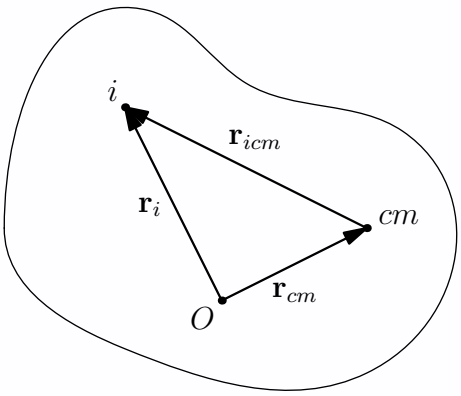


Fig. 1.11

By the definition of moment of inertia, the moment of inertia of the body around O is $I_O = \sum m_i r_i^2$. Writing r_i as a sum of vectors, we have $r_i = r_{icm} + r_{cm}$. Plugging into our definition yields

$$I_O = \sum m_i r_{icm}^2 + \sum m_i r_{cm}^2 + 2r_{cm} \cdot \sum m_i r_{icm}. \tag{1.60}$$

The first term is simply equal to I_{cm} by definition. The second term is equal to Mr_{cm}^2 since r_{cm} is a constant. The third term equals zero due to the definition of center of mass[7]. Thus, we have the parallel axis theorem, which states that the moment of inertia of a rigid body along an axis parallel to an axis through the center of mass equals the moment of inertia of the center of mass times Mr_{cm}^2 where r_{cm} is the distance between the axes. That is,

$$I_O = I_{cm} + Mr_{cm}^2 \tag{1.61}$$

★

[7]See sec. 1.3.5 for a more thorough explanation

1.4.4 ▶ ▶ ▶ Common Moment of Inertias

There are some common moment of inertias that should be memorized to increase speed while solving problems. However, it is also important to have the mathematical flexibility to be able to derive these expressions, especially for Physics C and above. All of these moment of inertias will be taken about an axis which passes through the center of mass[8], although one can find the moment of inertia about other axes with clever use of geometry paired with parallel and perpendicular axis theorem. To derive all of these, we will use the integral definition of moment of inertia, that is

$$I = \int r^2 dm. \tag{1.62}$$

Where the mass element dm is a distance r from the axis. To find dm, we consider a small portion of the object, then use the condition of uniform mass to get dm in terms of an integration variable. For more complicated shapes, we first find the moment of inertia of a small slice and integrate over the entire object.

▶ Starting with the simplest object, a ring of mass M and radius R. Start by considering a small section of the ring of length $Rd\theta$ with mass dm a distance R from the center. Using the condition of uniform mass density gives

$$\frac{dm}{Rd\theta} = \frac{M}{2\pi R}$$
$$dm = \frac{M}{2\pi} d\theta.$$

Plugging into the definition of moment of inertia and integrating yields

$$\begin{aligned} I &= \int_0^{2\pi} R^2 \frac{M}{2\pi} d\theta \\ &= \frac{MR^2}{2\pi} \cdot 2\pi \\ &= \boxed{MR^2} \end{aligned} \tag{1.63}$$

Note that this formula has no dependency on the height of the ring.

▶ Next is a disk of mass M and radius R. Consider a small ring of width dr and mass dm a distance r from the center of the disk. Using the condition of uniform mass density gives

$$\frac{dm}{2\pi r dr} = \frac{M}{\pi R^2}$$
$$dm = \frac{2M}{R^2} r dr.$$

Using eq. 1.62, the moment of inertia of this small ring is

$$dI = dmr^2 = \frac{2M}{R^2} r^3 dr.$$

[8] Which is also the geometrical center since we will only consider uniform mass here.

Integrating over the entire object gives

$$\begin{aligned} I &= \int_0^R \frac{2M}{R^2} r^3 dr \\ &= \frac{2M}{R^2} \cdot \frac{R^4}{4} \\ &= \boxed{\frac{1}{2} M R^2} \end{aligned} \tag{1.64}$$

Notice that this formula has no dependency on the height of the disk.

▶ Next is a rod of mass M and length L centered at the origin. Consider a small section of length dx and mass dm. Using the condition of uniform mass density gives

$$\begin{aligned} \frac{dm}{dx} &= \frac{M}{L} \\ dm &= \frac{M}{L} dx \end{aligned}$$

Using the definition of moment of inertia gives

$$\begin{aligned} I &= \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{M}{L} x^2 dx \\ &= \frac{M}{L} \cdot \frac{1}{3} \left(\frac{L^3}{8} + \frac{L^3}{8} \right) \\ &= \boxed{\frac{1}{12} M L^2} \end{aligned} \tag{1.65}$$

▶ Next is a spherical shell of mass M and radius R. To find the moment of inertia of this sphere, we can slice it into small rings at an angle of θ from the vertical and with width $Rd\theta$. These rings have mass dm and radius $R \sin\theta$.

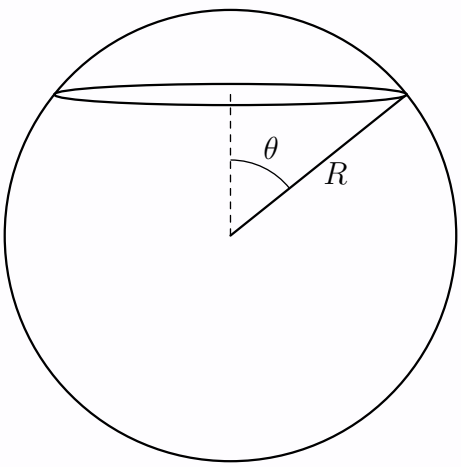


Fig. 1.12

Using the condition of uniform mass density gives

$$\frac{dm}{2\pi R\sin\theta \cdot Rd\theta} = \frac{M}{4\pi R^2}$$

$$dm = \frac{1}{2}M\sin\theta d\theta$$

Using our result from eq. 1.62, the moment of inertia for this small ring is

$$\begin{aligned} dI &= dm \cdot (R\sin\theta)^2 \\ &= \frac{1}{2}MR^2\sin^3\theta \end{aligned}$$

Integrating yields

$$\begin{aligned} I &= \frac{1}{2} \cdot \int_0^\pi \sin^3\theta d\theta \\ &= \frac{1}{2} \cdot \int_0^\pi \sin\theta(1-\cos^2\theta)d\theta \\ &= \frac{1}{2} \cdot \left[-\cos\theta + \frac{1}{3}\cos^3\theta\right]_0^\pi \\ &= \boxed{\frac{2}{3}MR^2} \end{aligned} \tag{1.66}$$

► Lastly, let us consider a solid sphere of mass M and radius R. Let us further consider small spherical shells of radius r and width dr. Using the condition of uniform mass density gives

$$\frac{dm}{4\pi r^2 dr} = \frac{M}{\frac{4}{3}\pi R^3}$$

$$dm = 3\frac{M}{R^3}r^2 dr$$

Using the result from eq. 1.65, the moment of inertia of each spherical shell dI is

$$dI = \frac{2}{3}dmr^2 = 2\frac{M}{R^3}r^4 dr.$$

Lastly, integrating yields

$$\begin{aligned} I &= 2\frac{M}{R^3}\int_0^R r^4 dr \\ &= \boxed{\frac{2}{5}MR^2} \end{aligned} \tag{1.67}$$

★

1.4.5 ▶ ▶ Perpendicular Axis Theorem

The perpendicular axis theorem provides an easy way to find moment of inertias perpendicular to two known axes, or for symmetrical shapes. Consider that we are trying to find the moment of inertia along the z axis I_z of an arbitrary object. This moment of inertia is

$$I_z = \int z^2 dm$$

where z is the distance of dm from the z axis. Using the Pythagorean Theorem, we have

$$\begin{aligned} I_z &= \int y^2 + x^2 dm \\ &= \boxed{I_x + I_y} \end{aligned} \tag{1.68}$$

Notice that $\int y^2 dm$ equals I_x and $\int x^2 dm$ equals I_y because y is the distance of dm from the x axis and vice versa ★

1.5 Gravitation

1.5.1 ▶ ▶ ▶ ▶ Gravitational Potential Energy

▶ The following formula for the gravitational potential of a system of two bodies is required for AP Physics 1, but the derivation is not required.

$$U = -\frac{Gm_1m_2}{r_{12}} \tag{1.69}$$

▶ ▶ ▶ To derive the above formula, we start with the fact that the gravitational force is a conservative force i.e. it can be written as the derivative of some scalar potential[9]

$$F = -\frac{dU}{dr} \tag{1.70}$$

The negative sign might not make sense at first but it is actually added as convention to give the equation more intuitive sense. The negative sign makes it so that conservative forces act in an attempt to decrease potential energy, which is what we experience in everyday life (if you walk up a hill, gravity tries to pull you down). Plugging Newton's Law of Gravitation into eq. 1.69 gives

$$-\frac{dU}{dr} = -\frac{Gm_1m_2}{r^2}$$

$$dU = \frac{Gm_1m_2}{r^2}dr$$

Now, to integrate, we must first define a point to zero potential, since the above expression gives no real indication of where zero is (the reason for +C in indefinite integrals). By convention, we set a point at infinity to zero, since it is common to all systems and is so far removed from all points in the system such that it is (usually) unaffected. However, one could theoretically choose any point they wish. Thus, integrating from infinity to the distance between the two bodies R gives

$$\int_0^{U(R)} dU = \int_\infty^R \frac{Gm_1m_2}{r^2}$$

$$U(R) = \left[-\frac{Gm_1m_2}{r}\right]_\infty^R$$

$$= -\frac{Gm_1m_2}{R} \tag{1.71}$$

which is the general form for gravitational potential energy. Now consider a mass m being raised a height h off the surface of a planet with radius R_E and mass M such that $h << R_E$. Using the above equation, its change in gravitational potential energy is

$$\Delta U = -\frac{GMm}{R_E} - \left(-\frac{GMm}{R_E + h}\right)$$

$$= GMm\left(\frac{1}{R_E + h} - \frac{1}{R_E}\right)$$

$$= GMm\left(\frac{R_E - (R_E + h)}{R_E(R_E + h)}\right)$$

$$= GMm\left(\frac{h/R_E^2}{1 + h/R_E}\right).$$

[9] See 1.3.1

Since $h << R_E$, the expression simply reduces to

$$\begin{aligned} \Delta U &= \frac{GMm}{R_E^2} h \\ &= mgh \end{aligned} \tag{1.72}$$

which is the approximation used for small displacements near the surface of a planet ★

1.5.2 ▶ ▶ Energy and Angular Momentum in Elliptical Orbits

In orbits, there are two important quantities that are conserved: energy and angular momentum. Since both of these quantities depend on the position of the satellite, by writing the quantities in terms of properties of the orbit, we can use their conservation to solve for the shape of the orbit. Before we begin, it would be helpful to review the properties of ellipses in A.2.2, because the following derivations will use them extensively.

To start, consider a satellite of mass m orbiting around a star of mass M in an elliptical orbit. Let the distance of closest approach, known as the perihelion distance, be r_p and let the distance of furthest approach, known as the aphelion distance, be r_a. At perihelion, the satellite has speed v_p while at aphelion it has speed v_a. By conservation of angular momentum,

$$\begin{aligned} mr_a v_a &= mr_p v_p \\ v_p &= \frac{r_a}{r_p} v_a \end{aligned} \tag{1.73}$$

Next, we can write conservation of energy for this satellite at perihelion and aphelion.

$$\frac{1}{2}mv_a^2 - \frac{GMm}{r_a} = \frac{1}{2}mv_p^2 - \frac{GMm}{r_p}$$

Rearranging, plugging v_p from eq. 1.72, and reducing with difference of squares yields

$$\begin{aligned} v_p^2 - v_a^2 &= 2GM\left(\frac{1}{r_p} - \frac{1}{r_a}\right) \\ v_a^2\left(\frac{r_a^2 - r_p^2}{r_p^2}\right) &= 2GM\left(\frac{r_a - r_p}{r_a r_p}\right) \\ v_a^2\left(\frac{r_a + r_p}{r_p}\right) &= 2GM\left(\frac{1}{r_a}\right) \\ v_a^2 &= \frac{r_p}{r_a}\frac{GM}{a} \end{aligned} \tag{1.74}$$

Now, we can find the angular momentum of the satellite, which is constant throughout the entire orbit. Plugging into the definition of angular momentum yields

$$\begin{aligned} L &= mr_a v_a \\ &= m\sqrt{r_a r_p}\sqrt{\frac{GM}{a}} \\ &= \boxed{mb\sqrt{\frac{GM}{a}}} \end{aligned} \tag{1.75}$$

Next, we can find the total energy of the system. At aphelion, the total energy is

$$E = \frac{1}{2}mv_a^2 - \frac{GMm}{r_a}.$$

Substituting v_a from eq. 1.73 gives

$$\begin{aligned} E &= \frac{1}{2}mv_a^2 - \frac{GMm}{r_a} \\ &= GMm\left(\frac{r_p}{2r_a a} - \frac{1}{r_a}\right) \\ &= GMm\left(\frac{r_p}{2r_a(r_a+r_p)} - \frac{r_a+r_p}{r_a(r_a+r_p)}\right) \\ &= GMm\left(\frac{-1}{r_a+r_p}\right) \\ &= \boxed{-\frac{GMm}{2a}} \end{aligned} \tag{1.76}$$

Using these forms of energy and angular momentum can help greatly in solving for the shape of orbits ★

1.5.3 ▶ ▶ ▶ Kepler's Second Law

Kepler's second law states that the area swept out by a satellite for any duration of time in its orbit is constant. Before I provide a derivation for it, it is important to note that all of Kepler's laws were discovered empirically by Johannes Kepler about ninety years before Newton published his law of universal gravitation. Thus, the following is more of a way to mathematically show that Kepler was correct in his laws, as opposed to a derivation which Kepler himself followed.

To start, consider a small amount of time Δt in which the satellite sweeps out an area ΔA (shown in grey) at a distance r from the star.

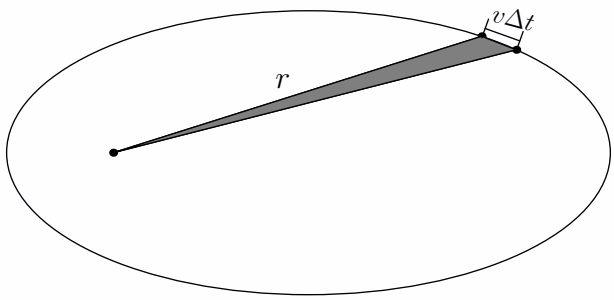


Fig. 1.13

If Δt is small enough, ΔA becomes a triangle[10] with base $v\Delta t$ and height r. Thus, its area is

$$\Delta A = \frac{1}{2} vr\Delta t$$

Rearranging and using the fact that $L = mrv$ gives

$$\frac{\Delta A}{\Delta t} = \frac{L}{2m} \tag{1.77}$$

which is a constant rate since L is a conserved quantity ★

[10] See A.2.1

1.5.4 ▶ ▶ ▶ Kepler's Third Law

Kepler's third law relates the period of a satellite's orbit to the semimajor axis of the orbit. To derive it, recall the statement of Kepler's second law, namely that the rate of change of area swept out in an orbit is constant with respect to time. Thus,

$$\frac{dA}{dt} = \frac{A}{T} \tag{1.78}$$

where A is the total area of the orbit and T is the period of the orbit. Plugging in the result from eq. 1.79 as well as the area of an ellipse gives

$$\frac{L}{2m} = \frac{\pi ab}{T}.$$

Lastly, plugging in out result from eq. 1.74 for L gives

$$\begin{aligned} T &= \frac{2m\pi ab}{L} \\ &= \frac{2m\pi ab}{mb\sqrt{\frac{GM}{a}}} \\ &= \boxed{\frac{2\pi a^{3/2}}{\sqrt{GM}}} \end{aligned} \tag{1.79}$$

★

1.5.5 ▶ ▶ ▶ Shell Theorem

Newton's law of universal gravitation provides an easy way to calculate the gravitational force between two point-like particles, but not for continuous bodies. Newton's Shell Theorem provides a way to calculate the gravitational field emanating from a spherical shell or solid sphere by integrating over small point-like masses, solving the issue of not knowing the gravitational field instated by continuous bodies.

To derive the Shell Theorem, consider a point at a distance D from a spherical shell of mass M and radius R. Furthermore, consider a slice of the shell as described by the following diagram

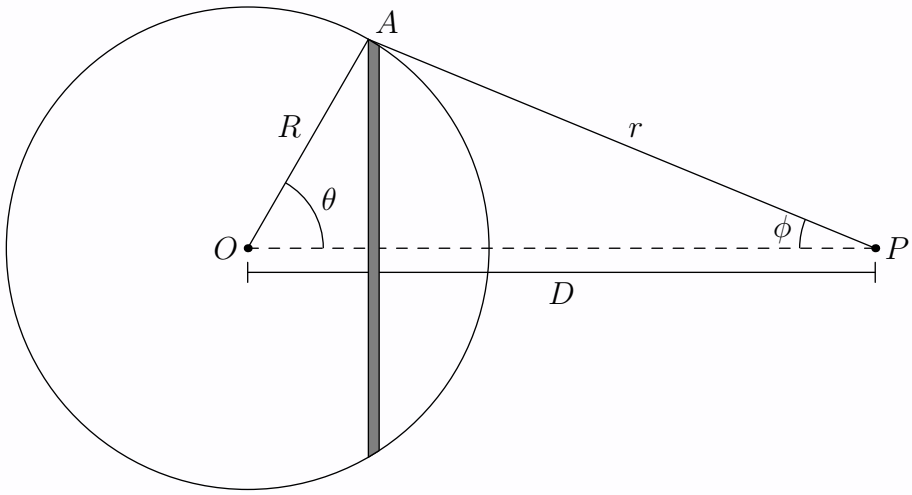


Fig. 1.14

At point P, the gravitational field due to the slice dg is given by

$$dg = G\frac{dm}{r^2}\cos\phi \tag{1.80}$$

where dm is the mass of the slice and only the horizontal component is taken since the vertical components from each point on the slice cancel out. The mass of the slice is given by

$$dm = \sigma 2\pi R \sin\theta R d\theta$$

where $\sigma = \frac{M}{4\pi R^2}$ is the surface mass density of the shell. Plugging values into eq. 1.76 and rearranging yields

$$dg = \frac{GM}{2}\cdot\frac{\sin\theta d\theta}{r^2}\cos\phi. \tag{1.81}$$

To simplify eq. 1.80, we can write two laws of cosines:

$$r^2 = R^2 + D^2 - 2RD\cos\theta$$

$$R^2 = r^2 + D^2 - 2rD\cos\phi$$

Differentiating the first one yields the relation

$$rdr = RD\sin\theta d\theta$$

Substituting all values into eq. 1.80 to write things in terms of r results in the expression

$$\begin{aligned} dg &= \frac{GM}{4RD^2} \cdot \frac{r^2 + D^2 - R^2}{r^2} dr \\ &= \frac{GM}{4RD^2} \left(1 + \frac{D^2 - R^2}{r^2}\right) dr \end{aligned} \tag{1.82}$$

Since r changes from $D - R$ to $D + R$ as it moves across the sphere, we integrate over those bounds

$$\begin{aligned} g &= \frac{GM}{4RD^2} \int_{D-R}^{D+R} 1 + \frac{D^2 - R^2}{r^2} dr \\ &= \frac{GM}{4RD^2} \left[r - \frac{D^2 - R^2}{r} \right]_{D-R}^{D+R} \\ &= \frac{GM}{4RD^2} \left[D + R - (D - R) - (D - R - (D + R)) \right] \\ &= \frac{GM}{4RD^2} [4R] = \boxed{\frac{GM}{4D^2}}. \end{aligned} \tag{1.83}$$

This result states that we can simply treat the spherical shell as a point mass located at the center of the sphere for points outside of the sphere. What if we were inside the sphere, though? Consider the following diagram, which is identical to Fig. 1.13 except point P is inside the shell.

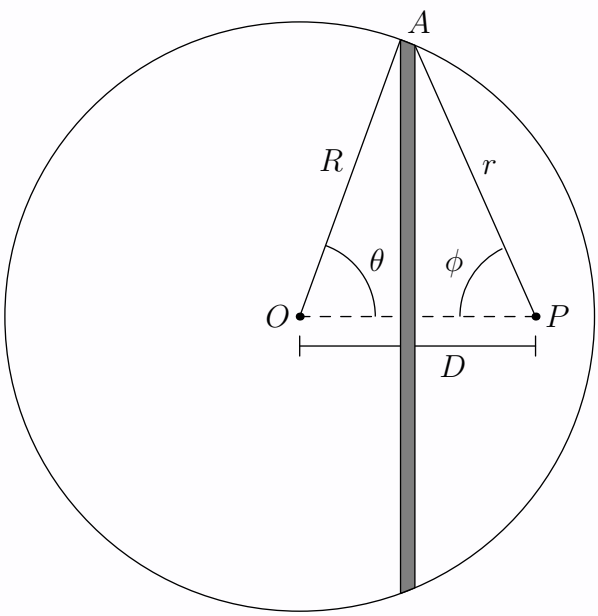


Fig. 1.15

In this case, the expression for the gravitational field at P is the same as the previous case, except the bounds of integration now go from $R - D$ to $R + D$. Calculating the

gravitational field at P using these new bounds gives us

$$\begin{aligned} g &= \frac{GM}{4RD^2}\left[r + \frac{R^2 - D^2}{r}\right]_{R-D}^{R+D} \\ &= \frac{GM}{4RD^2}\left[R - D + R + D - (R + D + R - D)\right] \\ &= \frac{GM}{4RD^2}\left[0\right] = \boxed{0}. \end{aligned} \tag{1.84}$$

This result that there is no gravitational field inside of a spherical shell can be surprising at first, but one can intuitively make sense of it by drawing a vertical line through P, and considering that although more mass lies to the left side of the line, it is on average farther away from P and vice versa[11].

Similar to the technique used in 1.4.3, we can now extend the result from eq. 1.83 to a solid sphere. For a point at a distance D from the center of a uniform solid sphere with radius R and mass M, the gravitational field due to a thin shell of the sphere is

$$dg = \frac{G\rho}{D^2}4\pi r^2 dr = \frac{3GM}{D^2R^3}r^2 dr.$$

Where $\rho = \frac{3M}{4\pi R^3}$ is the volume mass density of the sphere. Integrating over the entire sphere yields

$$g = \boxed{\frac{GM}{D^2}}, \tag{1.85}$$

which makes sense, since each shell of the sphere can be treated as a mass located at the center of the sphere.

To summarize, the gravitational field inside a spherical shell is zero, the gravitational field outside of solid spheres and spherical shells can be calculated by treating the sphere as a mass located at the center of the sphere ★

[11] Don't take this explanation too literally, it is just a way to make sense of the result!

1.6 Oscillations

1.6.1 ▶ ▶ ▶ Simple Harmonic Motion

Consider a particle moving along the x axis located at a position $x(t)$. By definition, simple harmonic motion is any motion in which the position of the particle is constrained by the following differential equation:

$$\ddot{x} = -\boldsymbol{\omega}^2 x \tag{1.86}$$

where $\boldsymbol{\omega}$ is an arbitrary constant. Typically, we write this equation as depicted above, but the constant could really be written anyway one wants. With this definition, we can solve the differential equation to find how the particle moves over time. Before we do that, though, let's classify this equation. We call this kind of differential equation a **homogeneous second order linear ordinary differential equation**. Here's what all of those words mean: **Homogeneous** meaning that all terms have $x(t)$ in them or a derivative of $x(t)$. **Second Order** meaning that the highest derivative is a second derivative. **Linear** meaning that the equation is linear in $x(t)$ and its derivatives. **Ordinary Differential Equation (ODE)** meaning that there are no partial derivatives. One of the most powerful techniques to solve homogeneous linear differential equations is to guess a solution then plug it into the equation and solve for constants. In this case, we start by guessing that the solution to the equation takes the form of

$$x(t) = Ae^{\alpha t}$$

where A and α are arbitrary constants. We choose to guess this exponential solution since the derivative of e^x is also e^x, so terms often cancel. Plugging this guess into eq. 1.77 yields

$$\alpha^2 Ae^{\alpha t} = -\boldsymbol{\omega}^2 Ae^{\alpha t}$$

$$\alpha = \pm i\boldsymbol{\omega}$$

Notice that we have two values for α, meaning that we will have two solutions to the differential equation. For linear differential equations, the most general solution to a differential equation is the sum of these solutions[12]. Thus, we have

$$x(t) = Ae^{i\boldsymbol{\omega}t} + Be^{-i\boldsymbol{\omega}t} \tag{1.87}$$

with A and B both being constants. Using Euler's identity, we have

$$\begin{aligned} x(t) &= A\cos\boldsymbol{\omega}t + iA\sin\boldsymbol{\omega}t + B\cos\boldsymbol{\omega}t - iB\sin\boldsymbol{\omega}t \\ &= \boxed{C\cos\boldsymbol{\omega}t + D\sin\boldsymbol{\omega}t} \qquad (1.88) \\ &= \boxed{E\cos(\boldsymbol{\omega}t + \phi)}. \qquad (1.89) \end{aligned}$$

The two boxed solutions to the differential equation are most commonly used, since they represent sinusoidal motion. Let us analyze each of them. In the first one (eq. 1.77), setting time to $t = 0$ gives $x(0) = C$, meaning that the constant C represents the initial position of the particle. If we take a derivative of the equation, then set $t = 0$, we have $\dot{x}(0) = \boldsymbol{\omega}D$, meaning that the constant D is related to the initial velocity of the particle.

[12] I won't prove this property of linear differential equations here, so you can just take this as a fact.

Thus, if we know the initial position and velocity of the particle, we should use 1.79 to represent the motion of the particle.

Looking at eq. 1.80, we see that the maximum value of the function is E. We call this value the amplitude of the motion[13]. Setting $t = 0$ in eq. 1.80 gives $x(0) = E\cos\phi$, meaning that ϕ has some relation to the starting position of the particle. We call this value the phase of the motion. Thus, we should use this form when we want to solve for the general properties of an oscillation[14] ★

[13] As a side note, $E = \sqrt{C^2 + D^2}$, which you should try to verify using addition of sinusoids and energy

[14] Notice how we assigned physical meaning to the constants C, D, E, and ϕ, but not to A and B. Why? Because A and B could be complex numbers and have no physical meaning

1.6.2 ▶ ▶ ▶ ▶ Period of a Spring Oscillator

The simplest example of simple harmonic motion is a block attached on a spring left to oscillate. If we consider a mass m attached to a spring of spring constant k, writing Hooke's Law with Newton's Second Law gives

$$ma = -kx$$

$$a = -\frac{k}{m}x \tag{1.90}$$

We know that if we have an equation in the form of eq. 1.89, the angular frequency of the oscillation is the square root of the coefficient in front of x (see sec. 1.6.1). Thus,

$$\boldsymbol{\omega} = \sqrt{\frac{k}{m}}$$

$$T = \boxed{2\pi\sqrt{\frac{m}{k}}} \tag{1.91}$$

Note the process taken to reach the result. We found an equation of motion that resembled that of simple harmonic motion. Then, we used the result of the solution to the equation of motion to find the frequency of the oscillation. This technique is extremely important when solving for the frequency of any simple harmonic oscillator ★

1.6.3 ▶ ▶ ▶ ▶ Period of a simple Pendulum

A pendulum is a good example of simple harmonic motion, and the periodic motion of its swinging has been used to keep time for centuries. To find the period of its oscillation, first consider a pendulum of length l with a mass m on the end. Furthermore, consider the pendulum at a small angle θ with respect to the vertical.

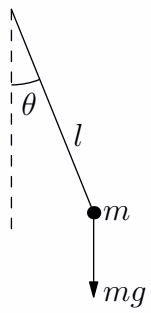


Fig. 1.16

Writing $\tau = I\alpha$ for this system about the pivot point gives

$$I\alpha = mgl\sin\theta$$

$$\alpha = \frac{mg}{ml^2}\sin\theta$$

$$\alpha = \frac{g}{l}\sin\theta \tag{1.92}$$

For small angles, $\sin\theta \approx \theta$[15]. So, we can write

$$\alpha = \frac{g}{l}\theta.$$

By the definition of SHM, we have

$$\boldsymbol{\omega} = \sqrt{\frac{g}{l}}$$

$$T = 2\pi\sqrt{\frac{l}{g}} \tag{1.93}$$

★

[15] See A.2.3

1.6.4 ▶ Damped Oscillations

In section 1.6.1, we looked at an idealistic version of a simple harmonic oscillator. In real life, though, oscillators will have some kind of resistive force acting on them. Consider a spring oscillator with mass m, spring constant k, and in a medium that provides a velocity-dependent resistive force of magnitude $F_d = bv$. Writing Newton's Second law for this system gives

$$m\ddot{x} = -b\dot{x} - kx$$

$$m\ddot{x} + b\dot{x} + kx = 0$$

Just like with the simple harmonic oscillator, this is a **homogeneous second order linear ordinary differential equation**. Thus, we employ the same method to solve it. Guessing a solution in the form of $x(t) = Ae^{\alpha t}$ and plugging in gives

$$m\alpha^2 + b\alpha + k = 0 \tag{1.94}$$

using the quadratic formula, we have

$$\begin{aligned}\alpha &= -\frac{b}{2m} \pm \frac{\sqrt{b^2 - 4mk}}{2m} \\ &= -\frac{b}{2m} \pm \sqrt{\frac{b^2}{4m^2} - \frac{k}{m}} \\ &= -\frac{1}{\tau} \pm \sqrt{\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2}. \end{aligned} \tag{1.95}$$

Where $\boldsymbol{\omega}_0 = \sqrt{\frac{k}{m}}$, and $\tau = \frac{2m}{b}$. Now, we can identify a few cases for the relative sizes of τ and $\boldsymbol{\omega}_0$.

Case 1: $\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2 > 0$
In this case, the solution to the differential equation is

$$x(t) = Ae^{t/\tau}\left(e^{\sqrt{\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2}\cdot t} + e^{-\sqrt{\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2}\cdot t}\right) \tag{1.96}$$

This case is known as overdamping, and it indicates that the oscillator goes back to the equilibrium position before completing one full oscillation.

Case 2: $\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2 = 0$
In this case, the solution is simply

$$x(t) = Ae^{-t/\tau} \tag{1.97}$$

Again, the oscillator returns to the equilibrium position before completing a full oscillation, but this time it reaches equilibrium in the fastest time possible. Thus, this case is known as critical damping.

Case 3: $\frac{1}{\tau^2} - \boldsymbol{\omega}_0^2 < 0$
In this case, we get an imaginary number in the exponent, so we can use Euler's identity to simplify, similar to the case for SHM. Thus, the solution to the differential equation is

$$x(t) = Ae^{-t/\tau}\cos(\boldsymbol{\omega}'t + \phi) \tag{1.98}$$

where $\boldsymbol{\omega}' = \sqrt{\boldsymbol{\omega}_0^2 - \frac{1}{\tau^2}}$ and ϕ is a constant. This case is known as underdamping, and the oscillator completes full oscillations, with its amplitude decaying to zero at infinity. Note that the frequency of the oscillation in this case doesn't actually change, and only the amplitude does. This is because the resistive force decreases the total energy of the system, which is dependent on the amplitude of the motion, and not the frequency.

1.6.5 ▶ Quality Factor

The quality factor or Q-Factor of a damped oscillator is a way of measuring how damped the oscillator is. A higher Q factor correlates with a higher quality oscillator (i.e. it isn't damped much) and vice versa. Mathematically, it is defined as follows:

$$Q = 2\pi \times \frac{E(t)}{E(t+T) - E(t)} \tag{1.99}$$

where $E(t)$ is the total energy of the system as a function of time and T is the period of oscillation. Consider a certain time when the oscillator comes to a stop. Further consider that the oscillator is at a position A at this time. The total energy of the system at this given moment is just the spring energy, or $\frac{1}{2}kA^2$. After one full oscillation takes place and the mass once again comes to rest, the position of the mass is $Ae^{-T/\tau}$ and the energy is $\frac{1}{2}kA^2e^{-2T/\tau}$. The energy lost in this cycle is $\frac{1}{2}kA^2\left(1 - e^{-2T/\tau}\right)$. Thus, the Q-Factor is

$$\begin{aligned} Q &= 2\pi \frac{\frac{1}{2}kA^2}{\frac{1}{2}kA^2\left(1 - e^{-2T/\tau}\right)} \\ &= \frac{2\pi}{1 - e^{-2T/\tau}} \end{aligned}$$

If we consider the case of minimal damping, $T \ll \tau$, we can use a first order Taylor approximation to simplify the exponential. Using $e^{-x} \approx 1 - x$, we have

$$\begin{aligned} Q &\approx \frac{2\pi}{1 - \left(1 - \frac{2T}{\tau}\right)} \\ &= \frac{\pi\tau}{T} \\ &= \frac{\pi\tau}{\frac{2\pi}{\omega}} \\ &= \boxed{\frac{\tau\omega}{2}} \end{aligned} \tag{1.100}$$

★

1.6.6 ▶ Driven Oscillations

In damped oscillators, the oscillation eventually dies out, meaning that an external force needs to be supplied to keep the system moving. This force should act with the movement of the object, so that it always is doing positive work on the object. For underdamped oscillators (the only case we will consider, since the other two damping cases don't oscillate), the frequency of this force should thus be the same as the frequency of the damped oscillator, and it should act in phase with the oscillator. Thus, the force must be in the form of

$$F_{driving} = F_0 cos(\boldsymbol{\omega} t) \tag{1.101}$$

where $\boldsymbol{\omega}$ is the natural or **resonant** frequency of the damped oscillator[16], and F_0 is a constant. Adding this driving force to the equation of motion for a damped oscillator yields the following differential equation:

$$m\ddot{x} + b\dot{x} + kx = F_0 cos(\boldsymbol{\omega} t) \tag{1.102}$$

This equation is an **inhomogeneous second order linear ODE**. To solve an inhomogenous ODE, we first find a particular solution to the equation (a couple methods to find one for this particular equation are listed below). Then, we add that particular solution to the solution of the homogeneous ODE[17]. This makes sense, since plugging the solution of the homogeneous equation into the ODE already gives zero, so linearly adding it to another term won't affect anything.

Method 1
Let the particular solution be the function $P(t)$. Plugging this function into eq. 1.91 gives

$$m\ddot{P}(t) + b\dot{P}(t) + kP(t) = F_0 \cos \boldsymbol{\omega} t \tag{1.103}$$

We see that the addition of a combination of different order derivatives of $P(t)$ yields a cosine function. Logically, then, $P(t)$ should be a sine or a cosine function, since they're each other's derivatives, and any sin or cosine function can be represented as a sum of other sine and cosine functions. Thus, we can confidently guess that the particular solution is of the form

$$P(t) = A \cos(\boldsymbol{\omega} t + \phi)$$

Where A and ϕ are constants we will solve for later.

Method 2
Let the particular solution be the function $P(t)$. Using the reasoning discussed above, the most general solution to the differential equation will look like

$$x(t) = A_0 e^{-t/\tau} \cos(\boldsymbol{\omega}' t + \phi_0) + P(t).$$

As time passes, the first term will decay to zero, leaving just the second term:

$$x(t)|_{t\to\infty} = P(t)$$

Now, we examine the purpose of the driving force in the first place. We added this force for the sole purpose of keeping the oscillator oscillating after a long time. Thus, we should

[16] We labeled this quantity as $\boldsymbol{\omega}'$ in 1.6.4
[17] i.e. setting the right hand side of eq. 1.91 to be zero.

expect that with this driving force, the oscillator continues to oscillate as time goes to infinity. Hence, the particular solution must just be the equation of motion of a simple harmonic oscillator i.e.

$$P(t) = A\cos(\omega t + \phi)$$

where A and ϕ are constants we will solve for later.

Now, we solve for A and ϕ. Plugging our particular solution into the differential yields

$$-m\omega^2 A\cos(\omega t + \phi) - b\omega A\sin(\omega t + \phi) + kA\cos(\omega t + \phi) = F_0\cos\omega t.$$

Expanding using additive trig identities yields the following massive equation that doesn't fit in a single line

$$-m\omega^2 A(\cos\omega t\cos\phi - \sin\omega t\sin\phi) - b\omega A(\sin\omega t\cos\phi + \cos\omega t\sin\phi)$$
$$+kA(\cos\omega t\cos\phi - \sin\omega t\sin\phi) = F_0\cos\omega t.$$

Factoring out the $\sin\omega t$ and $\cos\omega t$ terms yields

$$((kA - m\omega^2 A)\cos\phi - b\omega A\sin\phi)\cos\omega t$$
$$+((-kA + m\omega^2 A)\sin\phi - b\omega A\cos\phi)\sin\omega t = F_0\cos\omega t.$$

Since we know that the coefficients of the $\cos\omega t$ terms add up to F_0 and the coefficients of the $\sin\omega t$ terms add up to 0, we have the following system of equations

$$(kA - m\omega^2 A)\cos\phi - b\omega A\sin\phi = F_0 \tag{1.104}$$

$$(-kA + m\omega^2 A)\sin\phi - b\omega A\cos\phi = 0 \tag{1.105}$$

The full steps to solve this system will not be included here, since they are very messy. However, an overview will be given and the reader is encouraged to follow the general steps to solve the system. Solving for ϕ is easiest, and is simply done by rearranging the second equation to yield

$$\tan\phi = \frac{b\omega}{k - m\omega^2}. \tag{1.106}$$

Finding A requires rearranging the first equation and cleverly using the pythagorean identity $\tan^2\theta + 1 = \sec^2\theta$ to write everything in terms of $\tan\phi$. Then, plugging in $\tan\phi$ and reducing yields the result

$$A = \frac{F_0}{\sqrt{(k - m\omega^2)^2 - b^2\omega^2}} \tag{1.107}$$

★

1.7 Fluids

1.7.1 ▶ ▶ ▶ Hydrostatic Pressure

For ideal fluids, there is a direct correlation between pressure and depth within the fluid. To see this relation, consider a container of fluid with density ρ in hydrostatic equilibrium in an environment with ambient pressure P_0. Since no parts of the fluid are moving (at a macroscopic scale), all parts of the fluid must be in mechanical equilibrium. From this container, consider the cylinder of fluid with cross-sectional area A and height h outlined in the figure.

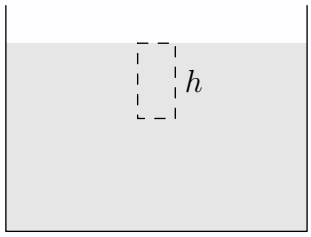


Fig. 1.17

There are three forces acting on this cylinder: an upward force from the hydrostatic pressure at depth h, a downward force from the ambient pressure in the environment, a downward force due to the weight of the fluid. Since the net force on the cylinder is zero, we have the following relation:

$$P(h)A = P_0A + mg.$$

Where $P(h)$ is the pressure of the fluid at depth h. Rewriting the mass of the cylinder using the density of the fluid yields

$$P(h)A = P_0A + \rho Ah \cdot g$$

$$\boxed{P(h) = P_0 + \rho gh} \tag{1.108}$$

Notice that this equation puts no constraint on the value or direction of g. So, if a fluid is placed in an accelerated reference frame where the g vector is no longer parallel, the pressure gradient will be anti-parallel to the direction of the g vector, and not necessarily vertical[18]. ★

[18] An argument can be made here that "vertical" is defined as the direction opposite to the g vector, but at the end of the day its quite semantical. Here we just consider a reference frame external to the fluid, where "vertical" isn't parallel to the effective gravity in the fluid.

1.7.2 ▶ ▶ ▶ Buoyant Force

Pressure increases with depth. Thus, if an object is submerged in a fluid, the bottom of the object experiences a greater pressure than the top, resulting in the force known as the buoyant force. Consider an object with mass m, height h, and cross-sectional area A submerged in a fluid of density $/rho$, and oriented such that the cross section is parallel to the surface of fluid. Further consider that the top of the object is a distance d from the surface of the fluid. Writing the force acting on the top and bottom of the object gives

$$F_{top} = AP_{top} = AP_0 + A\rho gd$$

$$F_{bottom} = AP_{bottom} = AP_0 + A\rho g(d+h)$$

Taking the difference of these forces gives

$$F_{net,pressure} = AP_0 + A\rho g(d+h) - AP_0 + A\rho gd = \rho gAh$$

Note that Ah is just the volume of the object, and thus

$$\boxed{F_{buoyant} = \rho gV} \tag{1.109}$$

Where V is the volume of the object. Notice therefore, that the buoyant force is equal to the weight of the water that is displaced by an object. If we further define ρ_w to be the density of the fluid and ρ_b to be the density of the object, the net force on the object is

$$F_{net} = (\rho_w - \rho_b)gV.$$

If the density of the object is greater than the fluid, it sinks and vice versa. If the density of the object and the fluid are equal, the object is suspended and doesn't accelerate. This effect is the reason why astronauts train on earth in large pools, where they specifically tune the average density of themselves to be equal to the water in the pool. This way, they can simulate the zero-g environment of outer space on earth.

▶ Notice that the above proof only accounts for objects that are an extrusion of a flat object (i.e. prisms), and a more general derivation requires more advanced calculus. To begin, first consider the following form of the net hydrostatic force exerted on an object submerged in a fluid

$$\sum F = \oiint_{dV} P \cdot dA. \tag{1.110}$$

This expression means sense, because it is essentially the sum of all hydrostatic force across the surface of the obect. Rewriting using the divergence theorem yields

$$\sum F = \iiint_V \nabla \cdot PdV \tag{1.111}$$

If we use the expression we found for pressure in 1.7.1, that is $\vec{P} = P_0 + \rho gz\hat{k}$, we can rewrite eq. 1.102 as

$$\begin{aligned}\sum F &= \iiint \rho gdV \\ &= \boxed{\rho gV}.\end{aligned} \tag{1.112}$$

★

1.7.3 ▶ ▶ ▶ Continuity Equation

The continuity equation represents the application of conservation of mass to fluids. Consider fluid flowing ideally through a section of a pipe with cross-section A_1 on the left side and A_2 on the right. The fluid flows at a speed v_1 on the side with area A_1 and vice versa. In a certain time Δt, a certain volume of fluid V_1 enters the tube on the left and a certain volume V_2 exits the tube on the right. The volume of water entering the tube is given by

$$V_1 = A_1 v_1 \Delta t$$

and the volume of water exiting the tube is given by

$$V_2 = A_2 v_2 \Delta t.$$

Since the fluid is incompressible and mass is conserved, the amount of fluid flowing in must be equal to the volume of water flowing out. Thus,

$$A_1 v_1 \Delta t$$

$$\boxed{A_1 v_1 = A_2 v_2} \tag{1.113}$$

The above equation is known as the continuity equation, and is the manifestation of conservation of mass in the flow of ideal fluids.

▶ For a more general equation of continuity that extends beyond fluid flow in a pipe, consider an arbitrary volume element with volume V and boundary dV. The net flux of fluid into the volume element is given by

$$\Phi_{net} = \oiint_{dV} \mathbf{u} \cdot d\mathbf{A}$$

where Φ_{net} is the net volume flow rate of fluid, $\mathbf{u}$ is the velocity field of the fluid, and $d\mathbf{A}$ is an area element on the surface dV. Since the fluid is incompressible and mass isn't created or destroyed, the net flux through the volume must be zero.

$$\oiint_{dV} \mathbf{u} \cdot d\mathbf{A} = 0 \tag{1.114}$$

Using the divergence theorem, we can alternatively write eq. 1.104 as

$$\boxed{\nabla \cdot \mathbf{u} = 0}. \tag{1.115}$$

★

1.7.4 ▶ ▶ ▶ Bernoulli's Equation

Bernoulli's equation is essentially just conservation of energy applied to a moving fluid. It gives an easy way to calculate properties of an ideal fluid at certain positions. To derive it, first consider the following diagram of an ideal fluid of density ρ in a tube.

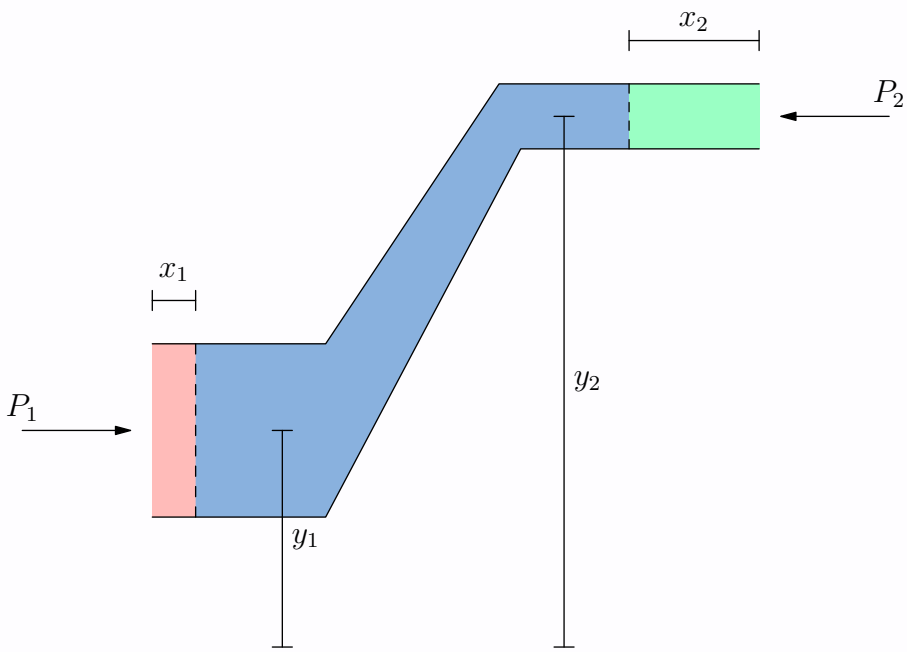


Fig. 1.18

Consider that the tube has cross-sectional area A_1 on the left end and cross-sectional area A_2 on the right. Further consider that the fluid has speed v_1 on the left end and v_2 on the right end. Imagine that fluid fills only the red and blue sections to begin. After some time, the fluid is pushed by the pressure differential created by P_1 and P_2, and the fluid advances to the blue and green sections. If we write the total energy of the fluid in its initial state, we have

$$E_i = \frac{1}{2}mv_1^2 + mgy_1 + E_b$$

where E_b is the total energy of the fluid in the blue section, and m is the mass of the fluid in the red section. After the fluid flows to the blue and green sections, the total energy of the fluid is

$$E_f = \frac{1}{2}mv_2^2 + mgy_2 + E_b.$$

Note that the mass of the fluid in the green section is equal to the mass of the fluid in the red section since the fluid is incompressible and mass is conserved. By the work-energy theorem, any changes in energy occur due to work done by net external forces. In this case, the work is done by the pressures P_1 and P_2. The total work done by these pressures is

$$W_{net} = P_1A_1x_1 - P_2A_2x_2.$$

Since we know that the mass of the fluid in the red green sections are equal, and the fluid

has a constant density, $A_1x_1 = A_2x_2 = \frac{m}{\rho}$, which means that

$$w_{net} = (P_1 - P_2)\frac{m}{\rho}.$$

From work energy theorem, we have

$$W_{net} = E_f - E_i$$

$$(P_1 - P_2)\frac{m}{\rho} = \frac{1}{2}mv_2^2 + mgy_2 - \frac{1}{2}mv_1^2 - mgy_1$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 \tag{1.116}$$

We can also write the above equation as

$$\boxed{P + \frac{1}{2}\rho v_2 + \rho g y = const} \tag{1.117}$$

which is Bernoulli's equation. Notice that if the height of the fluid doesn't change, pressure has an inverse relation to the kinetic energy. This inverse relation means that faster-moving fluid within a constant-energy stream has lower pressure than slower-moving fluid. This effect, known as the Bernoulli effect, is responsible for the lift force generated in sailboats and airplanes ★

Chapter 2

Thermodynamics

2.1 Statistical Thermodynamics

2.1.1 ▶ The Maxwell-Boltzmann Distribution

An ideal gas can be represented by a Gaussian distribution of the velocity of the singular particles in the gas, given by the following distribution:

$$f(v) = \frac{4}{\sqrt{\pi}} \left[\frac{m}{2k_B T}\right]^{3/2} v^2 e^{\frac{-mv^2}{2k_B T}} \tag{2.1}$$

The function is known as the **Maxwell-Boltzmann Distribution** and although it is a bit messy, it's important to recognize what it means.

The function is a **Probability Distribution Function (PDF)**, which is a special type of function that describes the way in which a random system is arranged. If we consider an arbitrary PDF $\rho(x)$ where x is some state of the system, ρ has no real meaning at any value of x. However, the area under ρ (integral) represents the probability of the sytem taking on a state that lies between the bounds of integration. Thus, any PDF $\rho(x)$ has the property

$$\int_{-\infty}^{\infty} \rho(x)dx = 1 \tag{2.2}$$

since the system must always be occupying one of its possible states. Similarly, the probability of an event occurring in the interval $(x, x + dx)$ is $\rho(x)dx$. For a container of gas, the particles are all evenly distributed, meaning that the probability of a gas particles having a velocity in the range $(v, v + dv)$ equals the fraction of all the particles with that velocity. Or,

$$\frac{dN}{N} = f(v)dv \tag{2.3}$$

where dN is the number of gas particles with a speed in the range $(v, v + dv)$. Recall that the expected value (or mean) of a system is defined as

$$\langle x \rangle = \sum_i x_i p(x_i)$$

where $\langle x \rangle$ is the average of x, x_i is an arbitrary state, and $p(x_i)$ is the probability of x_i occurring. It follows that for continuous distributions

$$\langle x \rangle = \int_{-\infty}^{\infty} x\rho(x)dx \tag{2.4}$$

since x is a state, $\rho(x)dx$ is the probability, and we are again summing over all possible states. Similarly, the mean of any function of x is

$$\langle g(x) \rangle = \int_{-\infty}^{\infty} g(x)\rho(x)dx.^{1} \tag{2.5}$$

[1] The reason why we don't change $\rho(x)$ to say, $g(\rho(x))$ is because the fundamental randomization process remains unchanged, and we are simply rescaling the quantity. For example, if we wanted to find the average of the square of a die roll, we would multiply the outcomes $(1, 4, 9, 16, 25, 36)$ by the probability $1/6$ and add them up to find the mean, which is the same probability used to calculate the mean value of the die normally. This all makes sense, because we aren't changing anything about the die roll, just the output value of each state. At any rate, these properties will be important when we examine macro-scale properties of systems of small particles.

Going back to the Maxwell-Boltzmann distribution, we can break it into two parts:

$$f(v) = \frac{4}{\sqrt{\pi}} \left[\frac{m}{2k_B T} \right]^{3/2}$$
$$\cdot\, v^2 e^{\frac{-mv^2}{2k_B T}}$$

The bottom part of the distribution relates to the actual distribution of the speeds of the particles in the gas[2] and is derived using statistical physics and experiment. The top part is a normalization factor, which gives the function the property from eq. 2.2. The Maxwell-Boltzmann distribution is special, though, since it's normalized in such a way that the area under the curve from 0 to ∞ is one, instead of the area along the entire x axis. To see why, consider the following graph of $f(v)$:

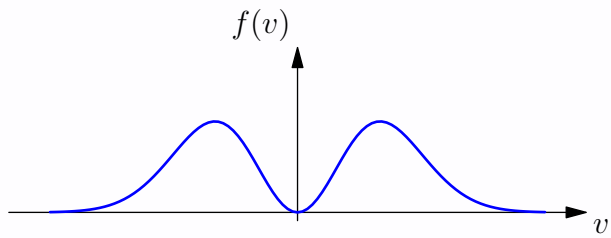


Fig. 2.1: Maxwell-Boltzmann Distribution

Since objects cannot have negative speed, all of the particles in a gas must lie on the right side of the graph, meaning

$$\int_0^\infty f(v)dv = 1. \tag{2.6}$$

This somewhat unusual normalization will be important in 2.1.2 when we integrate over all particles ★

[2] i.e. it's the part that gives the graph its shape

2.1.2 ▶ Root Mean Square Velocity

The root mean square (RMS) velocity is defined as the square root of the average of the squares of the velocities of the particles in a system. Mathematically (and possibly much more simply),

$$v_{RMS} = \sqrt{\langle v^2 \rangle}. \tag{2.7}$$

RMS quantities appear a lot in statistical physics, and they are used preferential to normal averages for a couple reasons:

1: The average value of an oscillating function may be zero, independent of the amplitude of the oscillations. However, it's hardly intuitive for a wildly-oscillating function to be described the same way as a barely-changing function. The RMS value fixes this discrepancy since it squares all values before taking an average, providing a more accurate representation of a quantity.

2: As we see in 2.1.3, since kinetic energy is defined with the square of the velocity, average velocity is useless if we want to calculate the average kinetic energy. Thus, v_{RMS} is used instead of the normal average.

In any case, since we know the PDF for particles in a gas, we can use the definitions listed in 2.1.1 to find a general expression. First, we can find the quantity $\langle v^2 \rangle$ by (notice the bounds of integration from eq. 2.6)

$$\langle v^2 \rangle = \int_0^\infty v^2 f(v) dv \tag{2.8}$$

where $f(v)$ is the Maxwell-Boltzmann distribution. Plugging in $f(v)$ and rearranging yields

$$\frac{2}{\sqrt{\pi}} a^{3/2} \int_0^\infty v^4 e^{-av^2} dv.$$

Where we use $a = m/2k_BT$. Integrating by parts with $u = v^3, du = 3v^2 dv$ and $dw = ve^{-av^2} dv, w = -\frac{1}{2a} e^{-av^2}$, we have

$$\frac{2}{\sqrt{\pi}} a^{3/2} \left[-\frac{v^3}{2a} e^{-av^2} \Big|_{-\infty}^{\infty} + \frac{3}{2a} \int_0^\infty v^2 e^{-av^2} dv \right] \tag{2.9}$$

The first term in the brackets is zero since the exponential decays to zero at infinity. To find the integral on the right, we could use integration by parts again, but instead we can use Feynman's trick for a quicker solution. By properties of the Gaussian integral [3],

$$\int_0^\infty e^{-av^2} dv = \frac{1}{2} \sqrt{\frac{\pi}{a}}.$$

Comparing this property with the integral in eq. 2.8, we can observe the following clever relation

$$\int_0^\infty v^2 e^{-av^2} dv = -\frac{d}{da} \int_0^\infty e^{-av^2} dv$$

[3] I won't prove it here, but the derivation is very interesting and you should definitely check it out!

where we take a derivative with respect to a, since the integral is a function of a [4]. Thus, the expression in eq. 2.8 becomes

$$\begin{aligned}\langle v^2 \rangle &= \frac{4}{\sqrt{\pi}} a^{3/2} \left[\frac{3}{2a} \left(-\frac{d}{da} \frac{1}{2} \sqrt{\frac{\pi}{a}} \right) \right] \\ &= \frac{4}{\sqrt{\pi}} a^{3/2} \left[\frac{3}{2a} \left(\frac{1}{4} \frac{\sqrt{\pi}}{a^{3/2}} \right) \right] \\ &= \frac{3}{2a}. \end{aligned} \tag{2.10}$$

The RMS speed is the square root of this value, or

$$\begin{aligned} v_{rms} &= \sqrt{\langle v^2 \rangle} \\ &= \sqrt{\frac{3}{2a}} \\ &= \sqrt{\frac{3k_B T}{m}} \end{aligned} \tag{2.11}$$

★

[4] v is a dummy variable, and is integrated out

2.1.3 ▶ ▶ Internal Energy

The total internal energy of a thermodynamic system is defined as the total energy contained within it. For an ideal gas, since there are no internal potentials, this energy is simply the total kinetic energy of the gas particles, or

$$U_{int} = N\langle K\rangle \tag{2.12}$$

where N is the number of particles and $\langle K\rangle$ is the number of gas particles. Using the mechanical definition of kinetic energy,

$$\langle K\rangle = \frac{1}{2}m\langle v^2\rangle = \frac{1}{2}mv_{rms}^2$$

Hence, the total internal energy of a monatomic gas is

$$U_{int} = \frac{3}{2}K_BT. \tag{2.13}$$

Alternatively, if we define n to be the number of moles of gas constant and the constant R to be Avogadro's number times k_B, we have

$$U_{int} = \frac{3}{2}(nN_A)k_BT = \frac{3}{2}nRT \tag{2.14}$$

Notice how this result is specifically for *monatomic* gas, and you can think of the equation as the sum of the average kinetic energy in each degree of freedom (x,y,z axis), which gives rise to the factor of 3/2.

$$K_i = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$$

For diatomic gasses, there are two[5] additional rotational degrees of freedom, namely

$$K_i = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{1}{2}I_x\boldsymbol{\omega}_x^2 + \frac{1}{2}I_y\boldsymbol{\omega}_y^2$$

Logically, the total internal energy should be

$$U_{int} = \frac{5}{2}Nk_BT \tag{2.15}$$

★

[5]Two, not three, since there is symmetry about two axes. For objects with less symmetry, there will be three rotational axes (Try it! Try to find more than two unique rotational axes for a pencil, then try the same with a smartphone!).

2.1.4 ▶ Ideal Gas Law

The famous Ideal Gas Law $PV = nRT$ was originally derived by experiment as an amalgamation of other emperical laws, namely Boyle's Law, Charles' Law, Gay-Lussac's Law, and Avogadro's Law. However, using the kinetic theory of gasses, it's possible to derive the law from first principles.

We start by considering a cube with side length L filled with a monatomic ideal gas. Since the gas exerts equal pressure in all directions, if we can relate the pressure of particles bouncing off of a single wall to the box's volume and temperature, we simply get the ideal gas law. For this derivation, we look at the particles bouncing off of the wall in the positive x axis (shown in blue). In a small time Δt, some particles traveling in the positive x direction collide with the blue surface, and those particles take up the volume defined by $\Delta l = \overline{v_x}\Delta t$ where $\overline{v_x}$ is the average x component of the velocities of the gas particles.

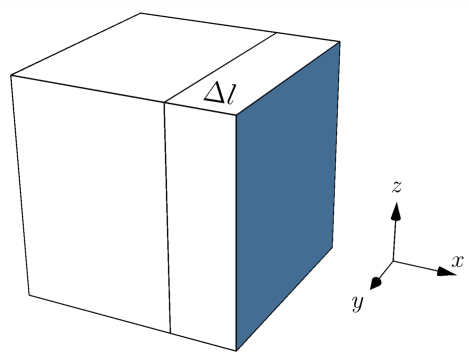


Fig. 2.2: Box with gas particles randomly moving inside. In a certain time Δt, some particles inside the section Δl hit the blue wall.

In this section of the cube, the number of particles is the number density of particles in the entire box multiplied by the volume of the section, since the particles are all evenly distributed. That number is

$$\frac{N}{V}\Delta l A = \frac{N A v_x \Delta t}{V}.$$

Of these particles that move on the x axis, half of them move towards the blue surface and half away. However, only those that move toward the surface exert any pressure, bringing our total considered particles to

$$\frac{N A v_x \Delta t}{2V}. \tag{2.16}$$

Since the gas is ideal, it collides elastically withe the walls of the container, imparting a force of $F = 2mv_x/\Delta t$ each. Thus, using eq. 2.16, the total force imparted on the blue side is

$$\frac{N A m v_x^2}{V}. \tag{2.17}$$

Since $P = F/A$, the average pressure on the blue side is

$$\langle P \rangle = \frac{N m \langle v_x^2 \rangle}{V}. \tag{2.18}$$

Since the particles are moving in all directions randomly, we can write the relation $v_{rms}^2 = \langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = 3\langle v_x^2 \rangle$, which implies that $\langle v_x^2 \rangle = \frac{k_B T}{m}$. Substituting everything in, we have[6]

$$P = \frac{N k_B T}{V}$$

$$PV = N k_B T \tag{2.19}$$

$$PV = nRT \tag{2.20}$$

★

[6] I removed the angle brackets around P here since the P in $PV = nRT$ represents the average pressure of the gas in a certain state of the gas. More specifically, the P in $PV = nRT$ represents the "average" over all of the microstates that a given macrostate encompasses.

2.2 Macroscopic Thermodynamics

2.2.1 ▶ ▶ Work Done on an Ideal Gas

Finding the work done on an ideal gas is important, since we can combine it with the first law of thermodynamics to derive some interesting relations (as seen later). To start, consider cylinder of gas of volume V topped with a movable piston of area A. If we apply a force F to the piston and depress it by a *distance* Δx, the work done by this force is simply

$$W = F\Delta x.$$

Dividing and multiplying by A yields

$$W = \frac{F}{A}A\Delta x$$

$$W = -P\Delta V. \tag{2.21}$$

Notice here that $A\Delta x = -\Delta V$. Since the cylinder is compressed, ΔV is negative. Thus, we need to add a negative sign to make things work out. This negative sign makes sense, since we'd expect that compressing a gas would do positive work. Why? Similar to a spring, when you compress the gas you work against a force, you're increasing the energy of the system. At a molecular level, you are simply increasing the momentum of the particles in the box as they collide with the surface of the piston! (see 2.1.4) ★

2.2.2 ▶ ▶ Heat Capacity of Gasses and Meyer's Relation

The specific heat capacity C of a substance is defined as the ratio of the amount of heat it takes to raise the temperature of a substance per mole of substance. In other words,

$$Q = nC\Delta T \tag{2.22}$$

Solids and liquids have relatively constant heat capacities over most temperature intervals, owing to their low thermal expansion. Gasses, on the other hand, have variable heat capacities, primarily because of their ability to change volume. Thus, it's helpful to define a few heat capacities for gasses, and we can even derive some useful relations between them.

First is the specific heat capacity at constant volume or C_v. Since the volume of the gas is constant, no work can be done on it and thus it can only change energy via heat transfer. Plugging in appropriate expressions into the First Law of Thermodynamics, we have (for a monatomic gas)

$$\begin{aligned} \Delta U &= Q \\ \frac{3}{2}nR\Delta T &= nC_v\Delta T \\ C_v &= \frac{3}{2}R \end{aligned} \tag{2.23}$$

The other notable heat capacity is the heat capacity at constant pressure. Writing out the First Law of Thermodynamics for a constant pressure pressure yields

$$\begin{aligned} \Delta U &= W + Q \\ \frac{3}{2}nR\Delta T &= -P\Delta V + nC_p\Delta T. \end{aligned} \tag{2.24}$$

Since pressure is constant, we have from the Ideal Gas Law that

$$\Delta V = \frac{nR\Delta T}{P} \tag{2.25}$$

Substituting eq. 2.25 into 2.24 and simplifying gives us

$$\begin{aligned} \frac{3}{2}nR\Delta T &= -nR\Delta T + nC_p\Delta T \\ C_p &= \frac{5}{2}R \\ C_p &= C_v + R. \end{aligned} \tag{2.26}$$

Equation 2.26 is known as Mayer's Relation and provides a convenient relation between the heat capacity of a gas at constant volume and pressure. An important thing to see is that it doesn't say anything about the values of C_v or C_p, just their relation. In fact, for diatomic gasses, the values for C_v and C_p are $5R/2$ and $7R/2$ respectively, which is a consequence of the phenomenon discussed towards the end of 2.1.3 ★

2.2.3 ▶ The Adiabatic Curve

On a P/V diagram, isobaric processes are a horizontal line, isochoric processes are a vertical line, and isothermal processes are a $1/x$ curve. These are all pretty intuitive, and can be derived in a few steps. Less intuitive is the curve for an adiabatic process, which we will derive below. By definition of an adiabatic process, any changes in internal energy of a system must arise only from work done on the system, or

$$dU = \frac{3}{2}nRdT = \delta W \tag{2.27}$$

where δ represents the a small change in a path dependent quantity. Using 2.2.1 and the Ideal Gas Law, we have

$$\begin{aligned}
\frac{3}{2}nRdT &= -PdV \\
nC_v dT &= -nR\frac{TdV}{V} \\
\frac{C_v}{R}\int \frac{dT}{T} &= -\int \frac{dV}{V} \\
\frac{C_v}{R}\ln T &= -\ln V + C
\end{aligned} \tag{2.28}$$

where C is some constant. Rewriting R in terms of C_p and C_v as well as defining the ratio $\gamma = C_p/C_v$, we have

$$\begin{aligned}
\ln T &= \frac{C_v - C_p}{C_v}\ln V + C \\
\ln T &= (1-\gamma)\ln V + C \\
TV^{\gamma-1} &= C
\end{aligned} \tag{2.29}$$

which itself is a perfectly good relation between quantities for an adiabatic process. However, since we're graphing things on a P/V diagram, it's nice to have a relation between P and V. Substituting the temperature using the Ideal Gas Law,

$$\begin{aligned}
\frac{PV}{nR}V^{\gamma-1} &= C \\
PV^{\gamma} &= const.
\end{aligned} \tag{2.30}$$

Since $\gamma > 1$, this curve looks similar to the curve for isothermal processes, but is a bit steeper ★

2.2.4 ▶ Carnot's Theorem

Carnot's theorem is a direct result of the second law of thermodynamics, and is an important result for engine design. It states that the most efficient heat engine that runs between two heat reservoirs is a perfectly reversible engine. The corollary of this statement is that all reversible engines running between two reservoirs of different temperature have the same efficiency.

We start the proof by considering a reversible heat engine R with efficiency η_R and an arbitrary heat engine E with efficiency η_E operating between a hot and cold reservoir.

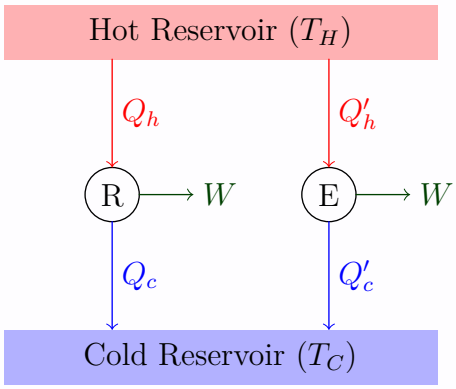


Fig. 2.3: A reversible and irreversible engine running between a hot and cold reservoir.

Both engines output work W but R takes in Q_h and outputs Q_c to T_C while E takes in Q'_h and outputs Q'_c to T_C. For the sake of the proof, assume that $\eta_E > \eta_R$, i.e.

$$\frac{W}{Q'_h} > \frac{W}{Q_h}$$

$$Q_h > Q'_h. \tag{2.31}$$

Now, consider running R in reverse using the work done by E, so that R extracts heat Q_c from T_C and deposits heat Q_h into T_H.

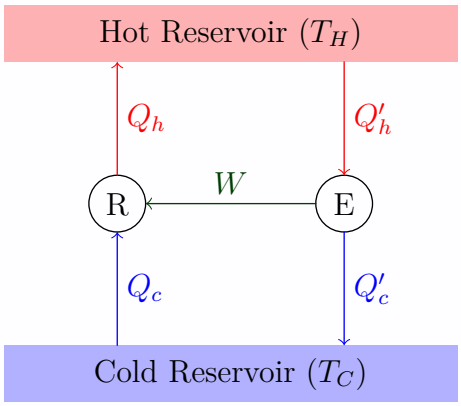


Fig. 2.4: A modified setup.

Analyzing the entire system at this state, the two engines extract heat $Q_c - Q'_c > 0$[7] from

[7] $Q_h = W + Q_c \Rightarrow$ e.q. 1.31 $\Rightarrow Q_c > Q'_c$

T_c and deposit heat $Q_h - Qh' > 0$ into T_h. Without any external work done on the system, this transfer of heat from a cold reservoir to a hot reservoir is a direct violation of the second law of thermodynamics, meaning that our initial assumption that $\eta_E > \eta_R$ must be false. Hence, there exists no engine more efficient than a reversible engine operating between two reservoirs.

If reversible engines are the most efficient engine between to temperature reservoirs, they must have the same efficiency, which is the maximum possible efficiency. Thus, the corollary mentioned earlier is also true. In real life applications, Carnot's theorem gives the upper bound on efficiency of a given heat engine, although most commercial engines don't come close to the ideal Carnot engine ★

2.2.5 ▶ Efficiency of a Carnot Engine

The Carnot Engine is a perfectly reversible engine operating between a hot and a cold temperature. As shown in 2.2.4, the efficiency of any such engine is the same for two reservoirs of given temperatures and is the upper bound for efficiency of an engine operating between the temperatures. A common way to construct the Carnot Engine is as follows (and as depicted in the diagram):

1. An isothermal expansion at hot temperature T_H
2. An adiabatic expansion
3. An isothermal compression at cold temperature T_C
4. An adiabatic compression

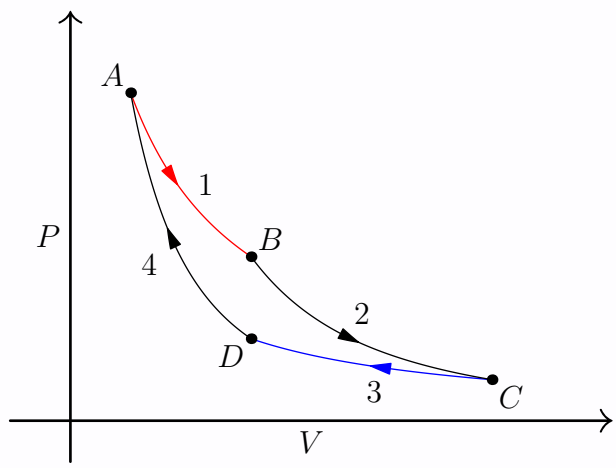


Fig. 2.5: Carnot Cycle, with two adiabatic and two isothermal processes. The hot temperature T_H is depicted in red while the cold temperature T_C is depicted in blue.

To find the efficiency of the engine, consider first the definition of engine efficiency, which is the ratio of the work output and the heat input. We can thus break each process down and see where heat is being inputted and where work is extracted (note that all of the following values are magnitudes).

1. Heat Q_H is inputted and work W_1 is extracted. Since $\Delta U = 0$, $Q_H = W_1$.[8]
2. No heat is exchanged but work W_2 is extracted.[9]
3. Heat Q_C is extracted and work W_3 is inputted. Since $\Delta U = 0$, $Q_C = W_3$.
4. No heat is exchanged but work W_4 is inputted.

[8]Note here that the volume increases at a positive pressure, meaning that the quantity $W = -P\Delta V$ is negative, but the work that the gas does is positive. Why? It's because the quantity $-P\Delta V$ is the work done *on* the gas, not *by* the gas, i.e. it gives the work inputted into the gas, not the work extracted. An intuitive way of thinking of it is like this: it takes effort to compress a piston and decrease its volume (positive work input) and if released from a compressed state, the piston springs outwards (positive work output).

[9]Again, W_2 is a positive value despite there being an increase in volume.

By conservation of energy, the work output of the engine is $Q_H - Q_C$, meaning that the efficiency of the engine is

$$\epsilon_{carnot} = \frac{W_{out}}{Q_H} = \frac{Q_H - Q_C}{Q_H}. \tag{2.32}$$

Since $Q_H = W_1$, we can find Q_H by solving for the area under process 1:

$$\begin{aligned} Q_H = W_1 &= \int_{V_A}^{V_B} PdV \\ &= nRT_H \int_{V_A}^{V_B} \frac{dV}{V} \\ &= nRT_H \ln\left(\frac{V_B}{V_A}\right) \end{aligned} \tag{2.33}$$

where V_A and V_B are the volumes of the gas at state A and B, respectively. By the same logic,

$$Q_C = W_3 = \int_{V_D}^{V_C} PdV = nRT_C \ln\left(\frac{V_C}{V_D}\right) \tag{2.34}$$

Plugging eq. 2.33 and 2.34 into 2.32, we have

$$\epsilon_{carnot} = 1 - \frac{T_C}{T_H} \cdot \frac{\ln(V_B/V_A)}{\ln(V_C/V_D)} \tag{2.35}$$

From the derivation in 2.2.3 (specifically eq. 2.29), we can relate the volumes between the start and end points of the adiabatic processes

$$\begin{aligned} \frac{T_H}{T_C} = \left(\frac{V_C}{V_B}\right)^{\gamma-1} &= \left(\frac{V_D}{V_A}\right)^{\gamma-1} \\ \frac{V_C}{V_D} &= \frac{V_B}{V_A}. \end{aligned}$$

With this result, eq. 2.35 simply reduces to

$$\boxed{\epsilon_{carnot} = 1 - \frac{T_C}{T_H}}. \tag{2.36}$$

★

Appendix A

Appendix

A.1 Useful Math References

A.1.1 Trigonometric Identities

$$\sin^2 x + \cos^2 x = 1$$
$$1 + \tan^2 x = \sec^2 x$$
$$1 + \cot^2 x = \csc^2 x$$

$$\sin(2x) = 2\sin x \cos x$$
$$\cos(2x) = \cos^2 x - \sin^2 x$$
$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

$$\sin\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{2}}$$
$$\cos\frac{x}{2} = \pm\sqrt{\frac{1+\cos x}{2}}$$
$$\tan\frac{x}{2} = \frac{1-\cos x}{\sin x} = \frac{\sin x}{1+\cos x}$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$
$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$
$$\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$
$$\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$

A.1.2 Taylor Series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\frac{1}{1+x} = 1 + x + x^2 + x^3 + \dots \{-1 < x < 1\}$$

A.2 Helpful Proofs

A.2.1 Arc length and a Straight Line

Consider an arc subtended by a very small angle $d\theta$ in a circle of radius R.

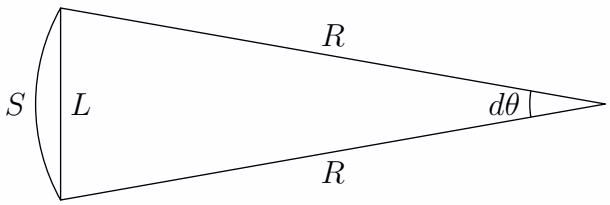


Fig. A.1

The distance S equals $R\theta$ while the distance L equals $2R\sin(d\theta/2)$. Taking the difference of these two lengths gives

$$\begin{aligned} S - L &= R\theta - 2R\sin\left(\frac{d\theta}{2}\right) \\ &= Rd\theta - 2R\left(\frac{d\theta}{2} - \frac{d\theta^3}{48} + \frac{d\theta^5}{3840} - \dots\right) \\ &\approx \frac{d\theta^3}{24} \end{aligned} \tag{A.1}$$

which is third-order small and thus will always be smaller than $d\theta$, which is already essentially zero. Since the two segments only differ to the third order in $d\theta$, we consider them to be equal for small angles ★

A.2.2 Properties of Ellipses

There is some important ellipse notation that should be understood before approaching elliptical orbits. Let us start with a diagram of an ellipse (note that it's not to scale):

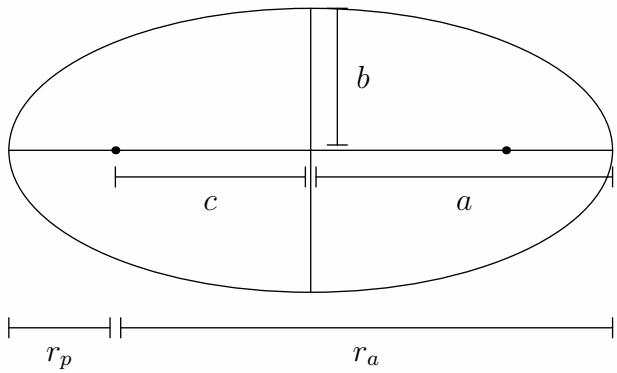


Fig. A.2

The distance a is the semi-major axis of the ellipse while the distance b is the semi-minor axis. The distance c is a representation of the eccentricity of the ellipse, i.e $c = \epsilon a$. If we place the star at the leftmost focus, r_p represents the perihelion distance while r_a represents the aphelion distance. From just these definitions, we can write

$$2a = r_a + r_p \tag{A.2}$$

$$2c = r_a - r_p \tag{A.3}$$

Next, by the definition of an ellipse, the distance from the top of the ellipse to each foci must be a. So, we can write

$$b^2 + c^2 = a^2 \tag{A.4}$$

Multiplying eq. A.4 through by 4 and substituting using A.2 and A.3 gives

$$4b^2 + r_a^2 - 2r_a r_p + r_p^2 = r_a^2 + 2r_a r_p + r_p^2$$

$$b = \sqrt{r_a r_p} \tag{A.5}$$

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A.2.3 Small Angle Approximations

For small angles, we often make a few approximations for trigonometric functions. $\sin(\theta)$ is approximated to θ, since writing out a taylor expansion yields

$$\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

Ignoring higher-order terms just yields $\sin\theta \approx \theta$. Doing the same for cosine yields

$$1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots$$

Ignoring high-order terms gives $\cos\theta \approx 1 - \frac{\theta^2}{2}$ ★